

22050 Continuous-Time Signals and Linear Systems

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L05

Fourier transformation

Fourier theorems

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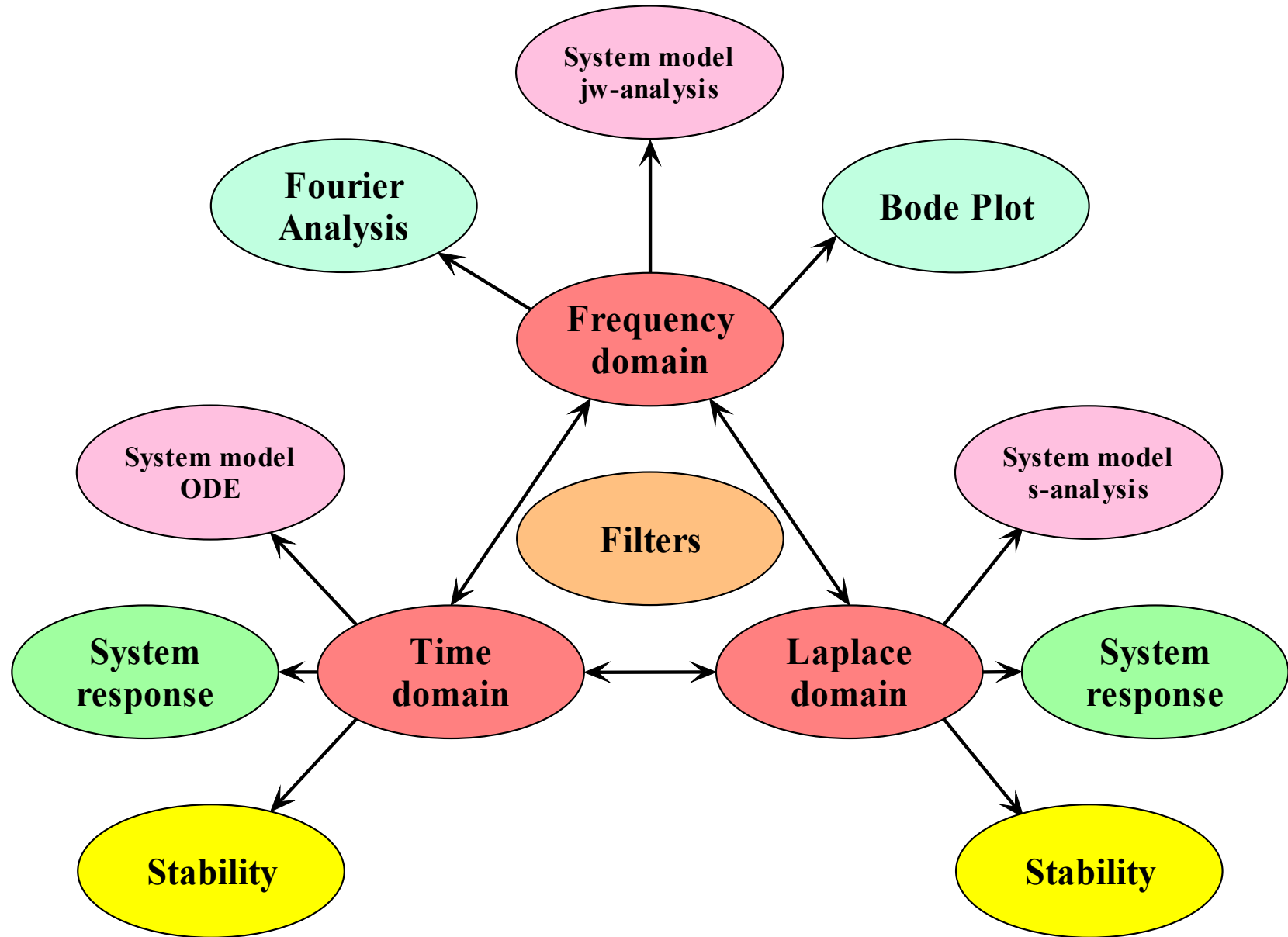
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- System
 - Model
 - Response
 - Performance
- Domains
 - Time
 - Frequency
 - s-domain



Lecture plan

- Aperiodic signals
 - Limitation of Fourier series
 - From Fourier series to Fourier transformation
 - Symmetry properties of the Fourier transformed
- Transformation of example signals
- Fourier Theorems
 - Time scaling
 - Time shift
 - Frequency shift and modulation
 - Convolution
 - Integration
 - Differentiation
- Distortion free systems
- The ideal versus the realistic filter

Lathi: Ch. 4.1 – 4.5 incl.

Truncated
exponential

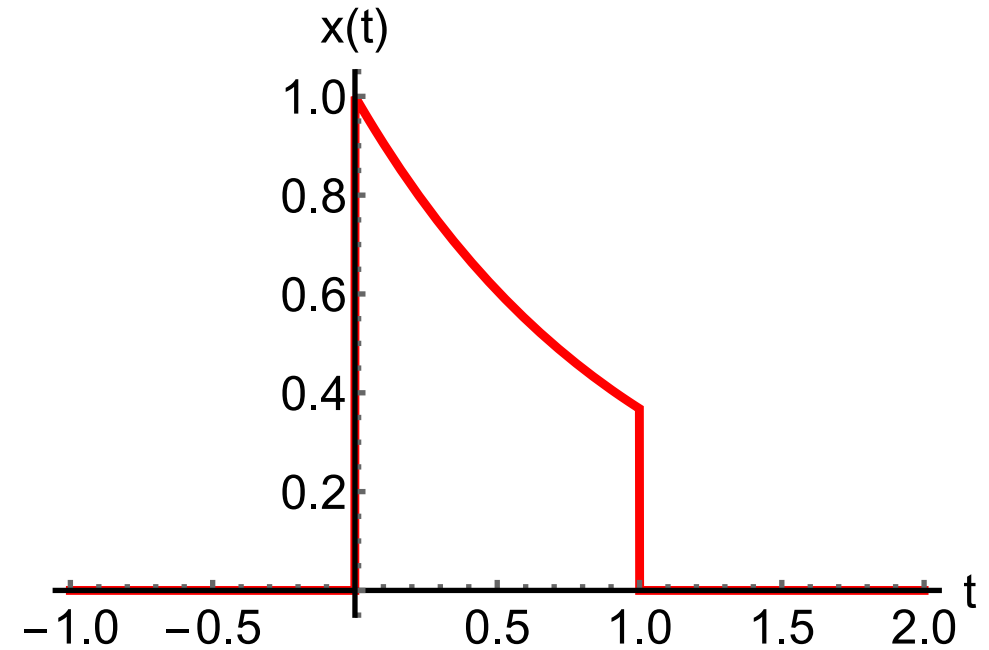
$$x(t) = \begin{cases} 0 & , \quad t < 0 \\ e^{-at} & , \quad 0 \leq t \leq T_0 \\ 0 & , \quad t > T_0 \end{cases}$$

$$f_0 = 1/T_0$$

What happens
to the Fourier
series coeffi-
cients if we let
 $T_0 \rightarrow \infty$?

$$\begin{aligned} D_n &= \frac{1}{T_0} \int_0^{T_0} e^{-at} e^{-j2\pi n f_0 t} dt \\ &= \frac{1}{T_0} \int_0^{T_0} e^{-(a+j2\pi n f_0)t} dt \\ &= \frac{1}{T_0} \frac{-1}{a+j2\pi n f_0} \left[e^{-(a+j2\pi n f_0)t} \right]_0^{T_0} \end{aligned}$$

$$\begin{aligned} D_n &= \frac{1}{T_0} \frac{1}{(a+j2\pi n f_0)} [1 - e^{-(a+j2\pi n f_0)T_0}] \\ &= \frac{1}{T_0} \frac{1}{(a+j2\pi n f_0)} [1 - e^{-aT_0} e^{-j2\pi n}] \\ &= \frac{1}{aT_0 + j2\pi n} [1 - e^{-aT_0}] \end{aligned}$$

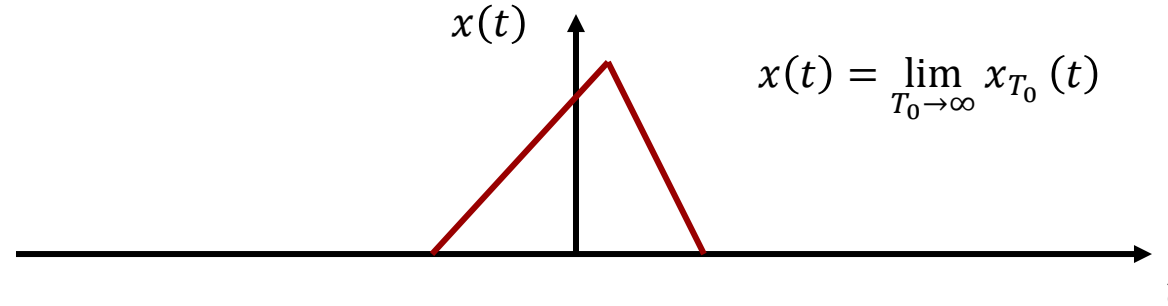


$$\lim_{T_0 \rightarrow \infty} |D_n| = \lim_{T_0 \rightarrow \infty} \frac{1 - e^{-aT_0}}{\sqrt{(aT_0)^2 + (2\pi n)^2}} = 0$$

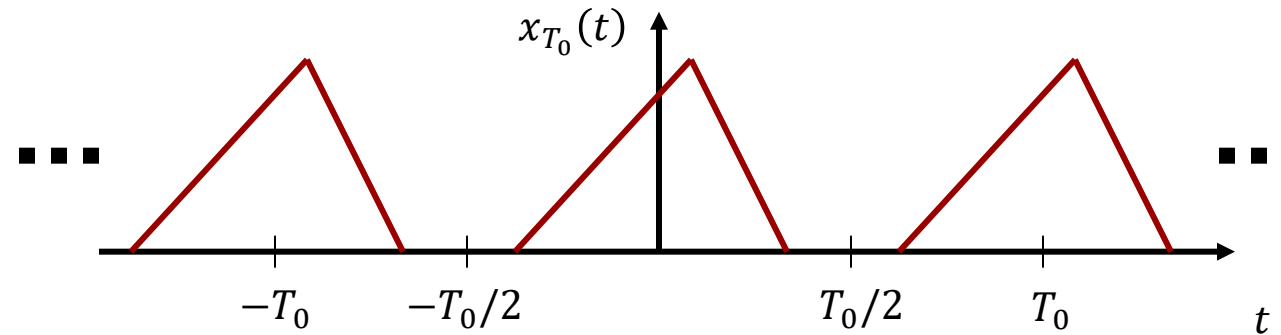
The Fourier Series cannot be used to analyze the spectra of aperiodic functions if they have infinite durations. This is usually the case with the impulse response.

Fourier Transformation

Aperiodic
signal:



Periodic
extension:

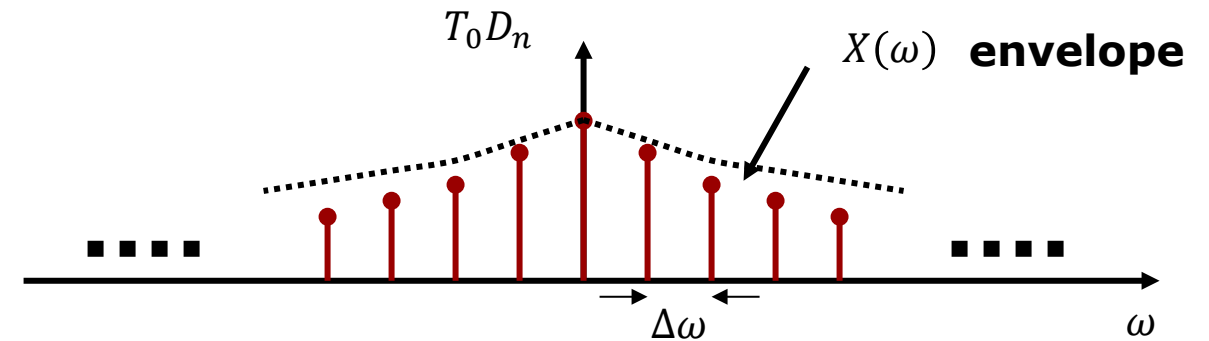


Integrating $x_{T_0}(t)$ from $-\frac{T_0}{2}$ to $\frac{T_0}{2}$
yields the same result as
integrating $x(t)$ from $-\infty$ to ∞ :

$$D_n = \frac{1}{T_0} \int_{-T_0/2}^{T_0/2} x_{T_0}(t) e^{-jn\omega_0 t} dt = \frac{1}{T_0} \int_{-\infty}^{\infty} x(t) e^{-jn\omega_0 t} dt$$

$$T_0 D_n = \int_{-\infty}^{\infty} x(t) e^{-jn\omega_0 t} dt$$

If we let $T_0 \rightarrow \infty$, we will get more and more stems placed closer and closer together. In the limit, we must be able to define the amplitude at any frequency, not just in discrete points on the frequency axis.



$$T_0 D_n = \int_{-\infty}^{\infty} x(t) e^{-jn\omega_0 t} dt = X(n\Delta\omega) \quad \omega_0 = \Delta\omega = \frac{2\pi}{T_0}$$

$$X(n\Delta\omega) = \int_{-\infty}^{\infty} x(t) e^{-jn\Delta\omega t} dt$$

$$X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt \quad \omega = \lim_{\Delta\omega \rightarrow 0} n\Delta\omega$$

Similarity of $x(t)$ with $e^{-j\omega t}$ for a continuous frequency spectrum

Fourier Synthesis:

$$\omega_0 = \Delta\omega = \frac{2\pi}{T_0}$$

$$\Downarrow$$

$$\frac{1}{T_0} = \frac{\Delta\omega}{2\pi}$$

We see that if the period $T_0 \rightarrow \infty$, then $\Delta\omega \rightarrow 0$.

In this limit, the sum becomes an integral and the discrete frequencies become a continuous frequency variable $n\Delta\omega \rightarrow \omega$:

$$x_{T_0}(t) = \sum_{n=-\infty}^{\infty} D_n e^{jn\omega_0 t} = \sum_{n=-\infty}^{\infty} \frac{X(n\Delta\omega)}{T_0} e^{jn\Delta\omega t}$$

$$= \frac{1}{2\pi} \sum_{n=-\infty}^{\infty} X(n\Delta\omega) e^{jn\Delta\omega t} \Delta\omega$$

$$x(t) = \lim_{T_0 \rightarrow \infty} x_{T_0}(t)$$

$$= \lim_{\Delta\omega \rightarrow 0} \frac{1}{2\pi} \sum_{n=-\infty}^{\infty} X(n\Delta\omega) e^{jn\Delta\omega t} \Delta\omega$$

$$= \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{j\omega t} d\omega$$

We have now established that an aperiodic signal can be represented by a continuous density function $X(\omega)$. We can think of $X(\omega)$ as the coefficients obtained by expanding the aperiodic signal into an infinite number of orthogonal basis functions the frequencies of which lies infinitely close ($\Delta\omega = 0$).

With this set of transforms we can go back and forth between time domain ($x(t)$) and frequency domain ($X(\omega)$).

We say that $X(\omega)$ is the Fourier transform of $x(t)$, and that $x(t)$ is the inverse Fourier transform of $X(\omega)$.

As $X(\omega)$ is a continuous density distribution of sinusoidal basis functions, $X(\omega)$ is also called a **frequency spectrum**.

The spectrum is a complex valued function of frequency with modulus and angle: $\angle$

Forward Transform:	$X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$	Analysis
--------------------	--	----------

Inverse Transform:	$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{j\omega t} d\omega$	Synthesis
--------------------	---	-----------

$X(\omega) = X(\omega) e^{j\angle X(\omega)}$	: Frequency spectrum
---	----------------------

$ X(\omega) $	: Amplitude spectrum
---------------	----------------------

$\angle X(\omega)$	: Phase spectrum
--------------------	------------------

$$X(\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t} dt$$

Substituting $-\omega$ for ω :

$$X(-\omega) = \int_{-\infty}^{\infty} x(t)e^{j\omega t} dt$$

Taking the complex conjugate:

$$X^*(-\omega) = \int_{-\infty}^{\infty} (x(t)e^{j\omega t})^* dt = \int_{-\infty}^{\infty} x^*(t)e^{-j\omega t} dt$$

If $x(t)$ is real then $x(t) = x^*(t) \Rightarrow$

$$X^*(-\omega) = \int_{-\infty}^{\infty} x^*(t)e^{-j\omega t} dt = \int_{-\infty}^{\infty} x(t)e^{-j\omega t} dt = X(\omega)$$

Conjugate symmetry property:

$$X(-\omega) = X^*(\omega)$$

$x(t)$ is real.

Conjugate symmetry property:

$$X(-\omega) = X^*(\omega)$$

$$X_{Re}(-\omega) + jX_{Im}(-\omega) = X_{Re}(\omega) - jX_{Im}(\omega)$$

Real part is even in ω :

$$X_{Re}(\omega) = X_{Re}(-\omega)$$

Imaginary part is odd in ω :

$$X_{Im}(-\omega) = -X_{Im}(\omega)$$

Modulus is even in ω :

$$|X(\omega)| = \sqrt{X_{Re}^2(\omega) + X_{Im}^2(\omega)} = \sqrt{X_{Re}^2(-\omega) + X_{Im}^2(-\omega)} = |X(-\omega)|$$

Angle is odd in ω :

$$\angle X(\omega) = \tan^{-1} \frac{X_{Im}(\omega)}{X_{Re}(\omega)} = \tan^{-1} \frac{-X_{Im}(-\omega)}{X_{Re}(-\omega)} = -\tan^{-1} \frac{X_{Im}(-\omega)}{X_{Re}(-\omega)} = -\angle X(-\omega)$$

We can also see the symmetry using an alternative approach:

Integrals of odd functions are zero.

The conjugate symmetry property:

$$\begin{aligned}
 X(\omega) &= \int_{-\infty}^{\infty} x(t)e^{-j\omega t} dt \\
 X(\omega) &= \int_{-\infty}^{\infty} (x_e(t) + x_o(t))(\cos \omega t + j \sin \omega t) dt \\
 &= \underbrace{\int_{-\infty}^{\infty} (x_e(t))(\cos \omega t) dt}_{\text{real part is even in } \omega} + \underbrace{\int_{-\infty}^{\infty} (x_e(t))(j \sin \omega t) dt}_{=0} \\
 &\quad + \underbrace{\int_{-\infty}^{\infty} (x_o(t))(\cos \omega t) dt}_{=0} + j \underbrace{\int_{-\infty}^{\infty} (x_o(t))(\sin \omega t) dt}_{\text{imaginary part is odd in } \omega}
 \end{aligned}$$

We now know what to expect when plotting the amplitude and phase spectra of a real-valued function:

Real-valued function $x(t)$:

$$\left. \begin{aligned} Re\{X(\omega)\} &= Re\{X(-\omega)\} \\ Im\{X(\omega)\} &= -Im\{X(-\omega)\} \end{aligned} \right\} \Rightarrow \begin{cases} |X(\omega)| = |X(-\omega)| \\ \angle X(\omega) = -\angle X(-\omega) \end{cases} \begin{array}{l} \text{Even function} \\ \text{Odd function} \end{array}$$

Fourier Transformation of Exponential function

Do all functions have a Fourier transform?

This example says no. It depends on the function.

$$x(t) = e^{-at} \cdot u(t), a > 0$$

$$X(\omega) = \int_{-\infty}^{\infty} e^{-at} u(t) e^{-j\omega t} dt$$

$$X(\omega) = \int_0^{\infty} e^{-at} e^{-j\omega t} dt = \int_0^{\infty} e^{-(a+j\omega)t} dt$$

$$X(\omega) = \frac{-1}{a+j\omega} \left[e^{-(a+j\omega)t} \right]_0^{\infty} = \frac{1}{a+j\omega} \left[1 - \lim_{T \rightarrow \infty} e^{-aT} e^{-j\omega T} \right]$$

Using: $|e^{-j\omega T}| = 1$

$$= \frac{1}{a+j\omega} \left[1 - \lim_{T \rightarrow \infty} e^{-aT} \right]$$

$$X(\omega) = \frac{1}{a+j\omega}, \text{ if } a > 0$$

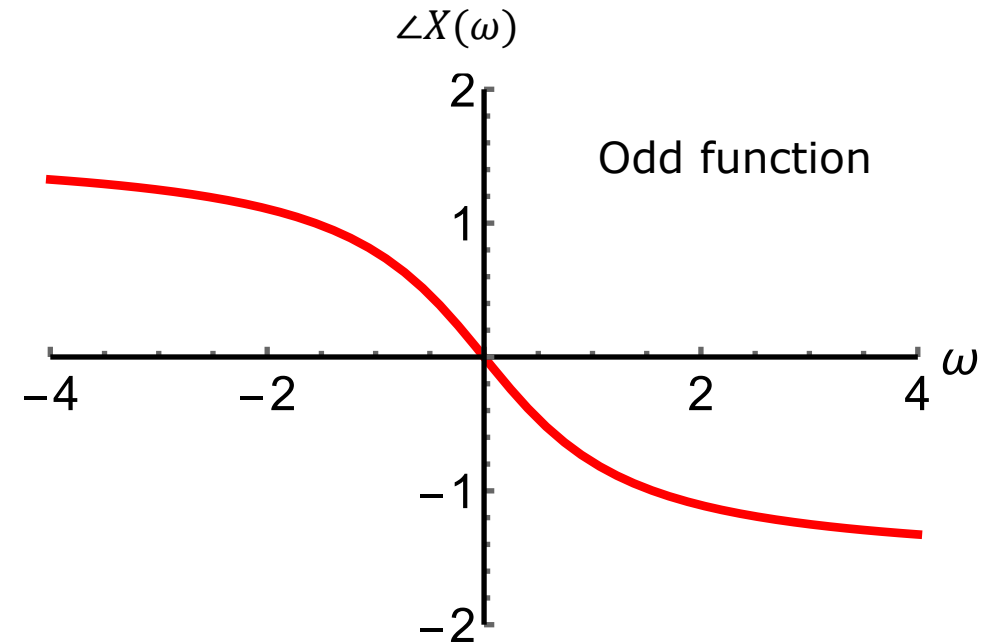
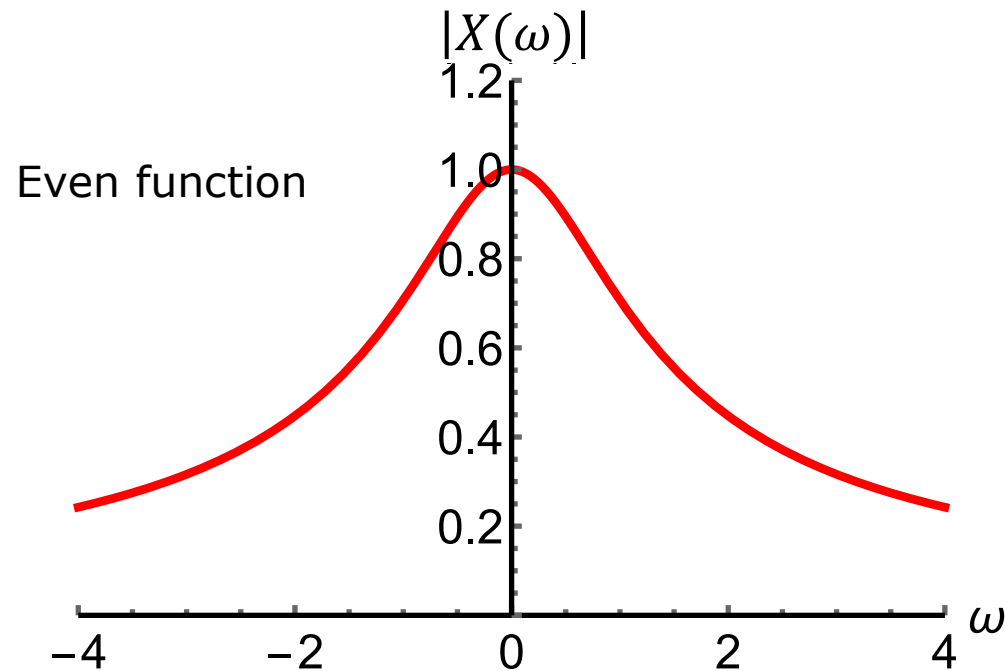
Observation:

The Fourier transform exists for decaying exponentials but not for growing exponentials.

Plotting Frequency Characteristic

Amplitude spectrum and phase spectrum for $a = 1$:

$$X(\omega) = \frac{1}{a + j\omega} = \frac{1}{\sqrt{a^2 + \omega^2}} \cdot e^{-j \tan^{-1}\left(\frac{\omega}{a}\right)}$$



$$\left. \begin{array}{l} \operatorname{Re}\{X(\omega)\} = \operatorname{Re}\{X(-\omega)\} \\ \operatorname{Im}\{X(\omega)\} = -\operatorname{Im}\{X(-\omega)\} \end{array} \right\} \Rightarrow \left\{ \begin{array}{ll} |X(\omega)| = |X(-\omega)| & \text{Even function} \\ \angle X(\omega) = -\angle X(-\omega) & \text{Odd function} \end{array} \right.$$

$$|X(\omega)| = \left| \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt \right| \leq \int_{-\infty}^{\infty} |x(t)| |e^{-j\omega t}| dt$$

$$|e^{-j\omega t}| = 1$$

$$|X(\omega)| \leq \int_{-\infty}^{\infty} |x(t)| dt < \infty$$

Dirichlet condition

A function must satisfy the Dirichlet condition to have a Fourier transform. In addition, it may only have a finite number of maxima and minima, and a finite number of jump discontinuities.

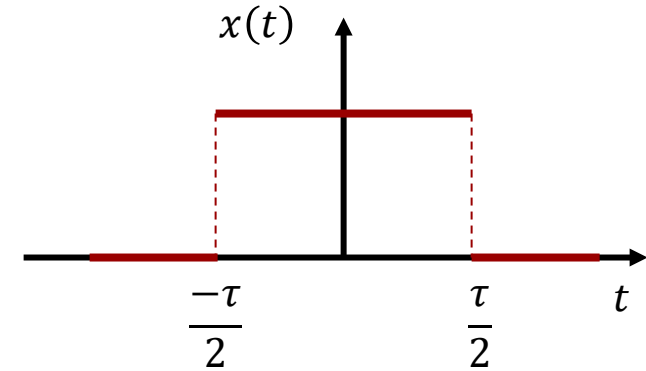
All physically possible signals obey the Dirichlet condition.

If the function $x(t)$ has jump discontinuities, the integral on the right will equal the average of the amplitudes approaching the jump from left and right.

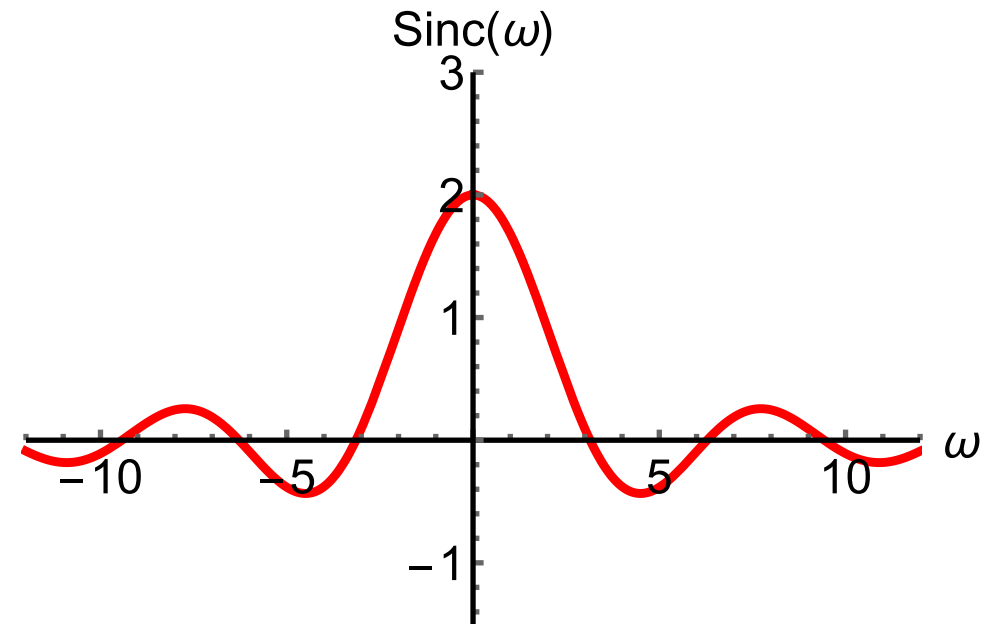
$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{j\omega t} d\omega$$

Pulse Function – Lathi: Ex. 4.2

$$x(t) = \text{rect}\left(\frac{t}{\tau}\right) = \begin{cases} 0 & , \quad t > |\tau/2| \\ 1/2 & , \quad t = |\tau/2| \\ 1 & , \quad t < |\tau/2| \end{cases}$$



$$\begin{aligned} X(\omega) &= \int_{-\tau/2}^{\tau/2} x(t) e^{-j\omega t} dt = \int_{-\tau/2}^{\tau/2} e^{-j\omega t} dt \\ &= \frac{-1}{j\omega} \left[e^{-j\omega t} \right]_{-\tau/2}^{\tau/2} = \frac{1}{j\omega} \left[e^{j\omega \tau/2} - e^{-j\omega \tau/2} \right] \\ &= \frac{2 \sin(\omega \tau/2)}{\omega} = \tau \frac{\sin(\omega \tau/2)}{\omega \tau/2} \\ &\stackrel{\text{def}}{=} \tau \text{sinc}(\omega \tau/2) \end{aligned}$$



For a square pulse of width τ ,
where are the zeros of
 $\text{sinc}(\omega \tau/2)$?

$$\begin{aligned} \text{sinc}(\omega_n \tau/2) &= 0 \\ \frac{\omega_n \tau}{2} &= n\pi \\ \omega_n &= \frac{n \cdot 2\pi}{\tau} \end{aligned}$$



```
restart
with(inttrans) :
with(DynamicSystems) :
with(plots) :
```

Useful functions:

```
dB := ω → 20 · log10(|H(ω)|) :           # converts to decibel
angle := ω → argument(H(ω)) ·  $\frac{180}{\pi}$  :   # converts from radians to degrees
```

Calculation of Fourier transform:

fourier(expr, t, w)

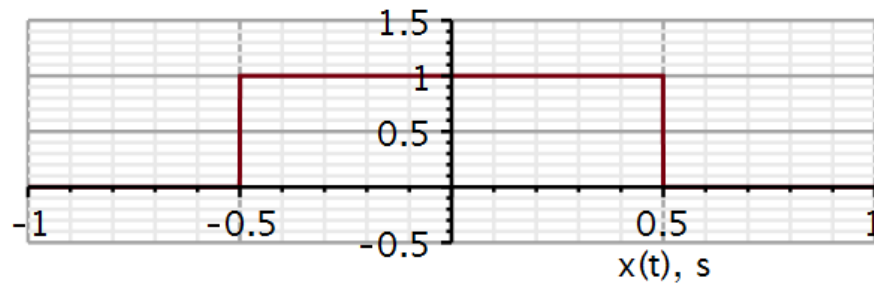
- expr - expression, equation, or set of equations and/or expressions to be transformed
- t - variable expr is transformed with respect to
- w - parameter of transform
- opt - option to run this under (optional)

Time domain signal

```
x := (Heaviside(t + 0.5) - Heaviside(t - 0.5)) :
```

Plotting signal

```
plot(x, t = -1 .. 1, -0.5 .. 1.5, thickness = 3, axesfont = [Helvetica, "roman", 18], axis[2] = [thickness = 2.5], axis[1] = [thickness = 2.5], labels = ["x(t), s", ""], labelfont = [Helvetica, 18], numpoints = 100, gridlines, size = [600, 200])
```

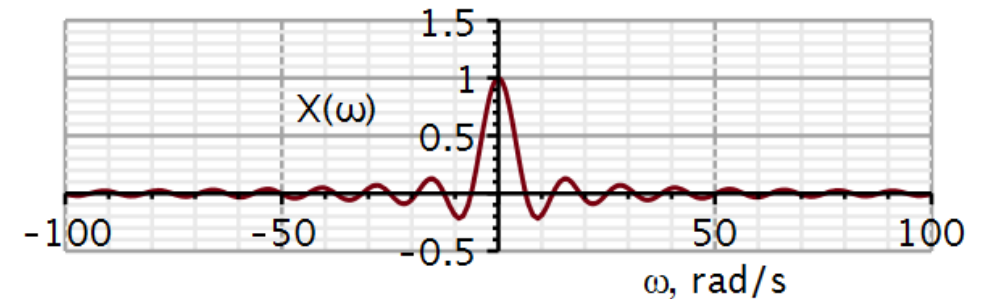


Calculating Fourier transform

```
X := fourier(x, t, ω)
```

$$X := \frac{2 \cdot \sin(0.5000000000 \omega)}{\omega}$$

```
plot(X, ω = -100 .. 100, -0.5 .. 1.5, thickness = 3, axesfont = [Helvetica, "roman", 18], axis[2] = [thickness = 2.5], axis[1] = [thickness = 2.5], labels = ["ω, rad/s", "X(ω)"], labelfont = [Helvetica, 18], numpoints = 100, gridlines, size = [600, 200])
```



In this case the Fourier transform is a real function. In general, it is complex-valued, and we need to plot the modulus and the angle in separate plots.

If I define a function as `x := t -> something` it does not do the Fourier transform.

Other Useful functions – Lathi (Ex. 4.3 – 4.6)

$$F\{\delta(t)\} = \int_{-\infty}^{\infty} \delta(t)e^{-j\omega t} dt = ? \quad 1$$

$$F^{-1}\{\delta(\omega)\} = \frac{1}{2\pi} \int_{-\infty}^{\infty} \delta(\omega)e^{j\omega t} d\omega = ? \quad \frac{1}{2\pi}$$

$$F^{-1}\{2\pi\delta(\omega)\} = \int_{-\infty}^{\infty} \delta(\omega)e^{j\omega t} d\omega = ? \quad 1$$

$$F^{-1}\{2\pi\delta(\omega - \omega_0)\} = \int_{-\infty}^{\infty} \delta(\omega - \omega_0)e^{j\omega t} d\omega = ? \quad e^{j\omega_0 t}$$

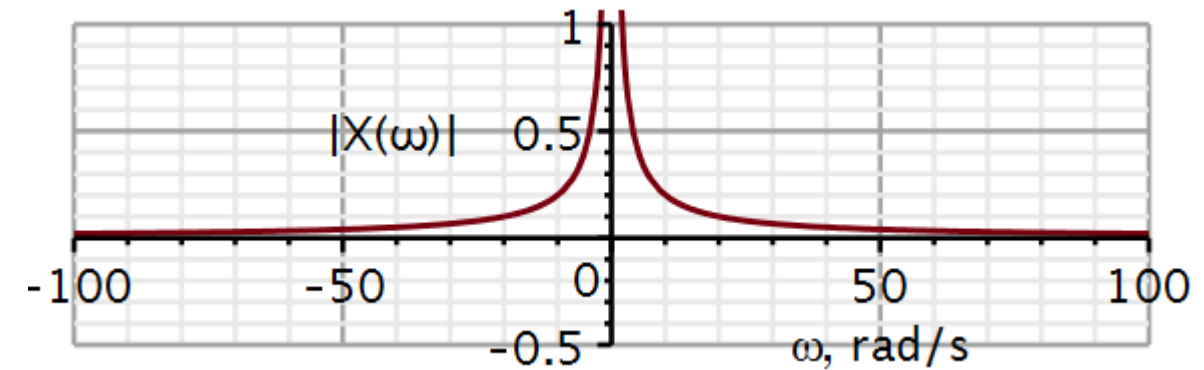
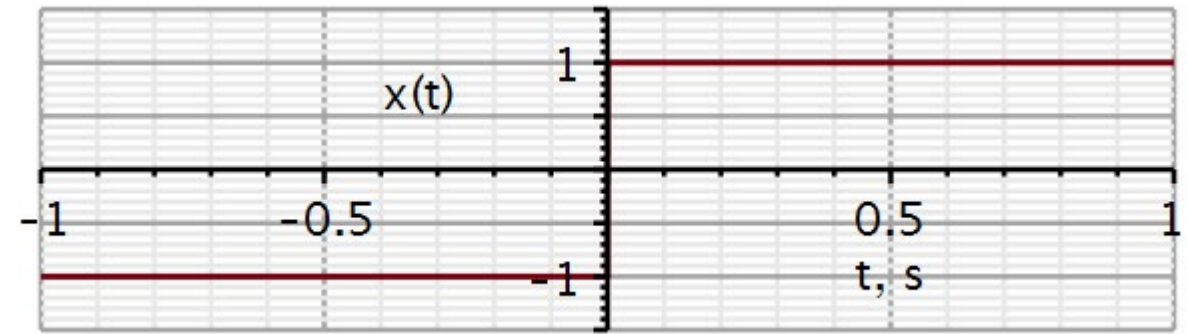
$$F^{-1}\{\pi(\delta(\omega - \omega_0) + \delta(\omega + \omega_0))\} = ? \quad \frac{e^{j\omega_0 t} + e^{-j\omega_0 t}}{2} = \cos(\omega_0 t)$$

$$\text{sgn}(t) = \begin{cases} -1 & t < 0 \\ 1 & t > 0 \end{cases} = \lim_{a \rightarrow 0} [e^{-at}u(t) - e^{-a(-t)}u(-t)]$$

$$F\{e^{-at}u(t)\} = \frac{1}{a + j\omega}$$

$$\begin{aligned} \int_{-\infty}^{\infty} e^{at}u(-t)e^{-j\omega t}dt &= \int_{-\infty}^0 e^{(a-j\omega)t}dt \\ &= \frac{1}{a-j\omega} [e^{(a-j\omega)t}]_{-\infty}^0 = \frac{1}{a-j\omega} \end{aligned}$$

$$F\{\text{sgn}(t)\} = \lim_{a \rightarrow 0} \left[\frac{1}{a + j\omega} - \frac{1}{a - j\omega} \right] = \frac{2}{j\omega}$$

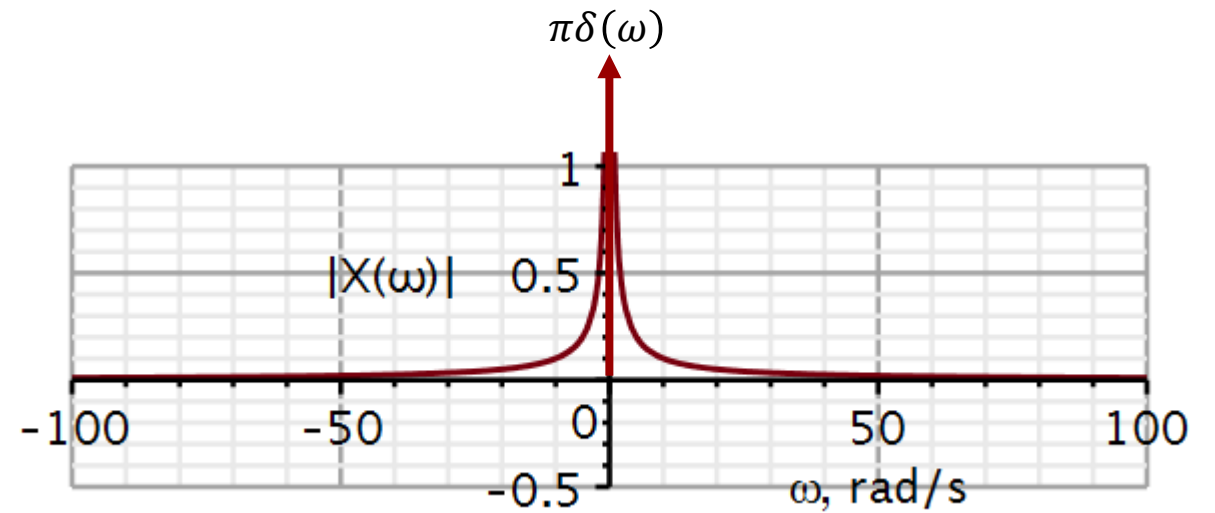
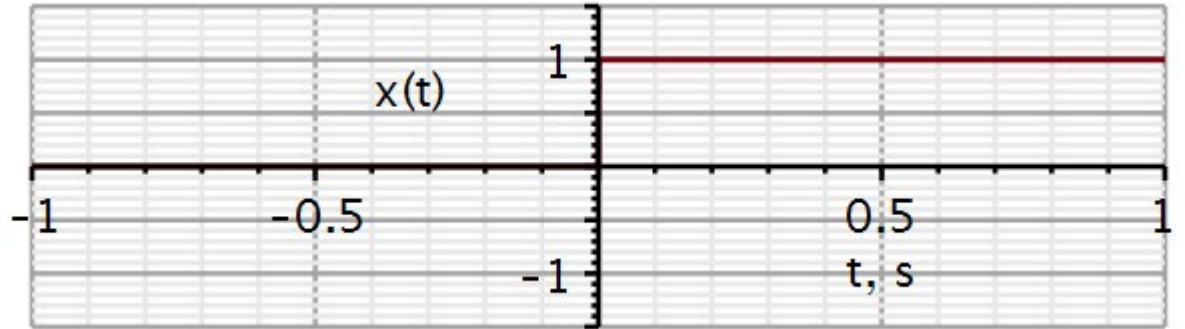


Step function – Lathi (Ex. 4.9)

$$u(t) = \frac{1}{2} + \frac{1}{2} \text{sgn}(t)$$

$$F\{u(t)\} = \frac{1}{2} F\{1\} + \frac{1}{2} F\{\text{sgn}(t)\}$$

$$\begin{aligned} F\{u(t)\} &= \frac{1}{2} \cdot 2\pi\delta(\omega) + \frac{1}{2} \cdot \frac{2}{j\omega} \\ &= \pi\delta(\omega) + \frac{1}{j\omega} \end{aligned}$$



Fourier theorems

Lathi 4.3

Video 2

If we have a known function $x(t)$, we can calculate its Fourier transform $X(\omega)$. This way we have established a transform relationship between two function.

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{j\omega t} d\omega$$

$$X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

$$x(t) \leftrightarrow X(\omega)$$

We have obtained a function X . What would be the Fourier transform of this function, if we know consider this function to be a function of time?

In other words, what is $\mathcal{F}\{X(t)\}$?

First, we calculate the Fourier transform:

$$X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt = \int_{-\infty}^{\infty} x(s) e^{-j\omega s} ds$$

Now we use that same function as a time function. Using variable substitution, we have an expression for $X(t)$:

$$X(t) \stackrel{\text{def}}{=} \int_{-\infty}^{\infty} x(s) e^{-jst} ds$$

We can calculate the Fourier transform of this new time function:

$$\mathcal{F}\{X(t)\} = \mathcal{F}\left\{ \int_{-\infty}^{\infty} x(s) e^{-jst} ds \right\} = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x(s) e^{-jst} ds e^{-j\omega t} dt$$

Changing order of integrals:

$$\mathcal{F}\{X(t)\} = \int_{-\infty}^{\infty} x(s) \int_{-\infty}^{\infty} e^{-j(s+\omega)t} dt ds$$

We need to figure out the red integral.

The inverse Fourier transform of $\delta(\omega)$ is:

It follows that the forward Fourier integral of 1 is:

We can now compute the red integral:

$$\mathcal{F}^{-1}\{2\pi\delta(\omega)\} = \int_{-\infty}^{\infty} \delta(\omega) e^{j\omega t} d\omega = 1$$

$$\mathcal{F}\{1\} = \int_{-\infty}^{\infty} e^{-j\omega t} dt = 2\pi\delta(\omega)$$

$$\int_{-\infty}^{\infty} e^{-j(s+\omega)t} dt = 2\pi\delta(s + \omega)$$

$$\mathcal{F}\{X(t)\} = \int_{-\infty}^{\infty} x(s) \int_{-\infty}^{\infty} e^{-j(s+\omega)t} dt ds$$

$$= \int_{-\infty}^{\infty} x(s) 2\pi\delta(s + \omega) ds = 2\pi x(-\omega)$$

If $X(\omega)$ is the forward Fourier transform of $x(t)$ then

$2\pi x(-\omega)$ is the forward Fourier transform of $X(t)$

$$2\pi x(t) = \int_{-\infty}^{\infty} X(\omega) e^{j\omega t} d\omega \quad x(t) \leftrightarrow X(\omega)$$

Exchange t with $-t$

$$2\pi x(-t) = \int_{-\infty}^{\infty} X(\omega) e^{-j\omega t} d\omega$$

Exchange t and ω

$$2\pi x(-\omega) = \int_{-\infty}^{\infty} X(t) e^{-j\omega t} dt \quad X(t) \leftrightarrow 2\pi x(-\omega)$$

If $X(\omega)$ is the forward Fourier transform of $x(t)$ then
 $x(-\omega)$ is the forward Fourier transform of $X(t)$

We have calculated the transform fair

$$x(t) = \text{rect}\left(\frac{t}{\tau}\right) \leftrightarrow \tau \text{ sinc}\left(\frac{\tau}{2} \omega\right) = X(\omega)$$

Using the Duality property:

$$X(t) \leftrightarrow 2\pi x(-\omega)$$

We obtain:

$$\tau \text{ sinc}\left(\frac{\tau}{2} t\right) \leftrightarrow 2\pi \text{ rect}\left(\frac{\omega}{\tau}\right)$$

We can check by doing the math:

$$X(\omega) = 2\pi \text{ rect}\left(\frac{\omega}{\tau}\right) = 2\pi \begin{cases} 0 & , \quad \omega > |\tau/2| \\ 1/2 & , \quad \omega = |\tau/2| \\ 1 & , \quad \omega < |\tau/2| \end{cases}$$

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{j\omega t} d\omega$$

$$x(t) = \frac{1}{2\pi} \int_{-\tau/2}^{\tau/2} 2\pi e^{j\omega t} d\omega = \frac{1}{jt} [e^{j\omega t}]_{-\tau/2}^{\tau/2} = \frac{e^{j\frac{\tau}{2}t} - e^{-j\frac{\tau}{2}t}}{jt}$$

$$x(t) = \frac{2 \sin\left(\frac{\tau}{2} t\right)}{t} = \tau \frac{\sin\left(\frac{\tau}{2} t\right)}{\frac{\tau}{2} t} = \tau \text{ sinc}\left(\frac{\tau}{2} t\right)$$

Using variable
substitution:

$$a > 0$$

$$s \stackrel{\text{def}}{=} at$$

$$\frac{ds}{a} = dt$$

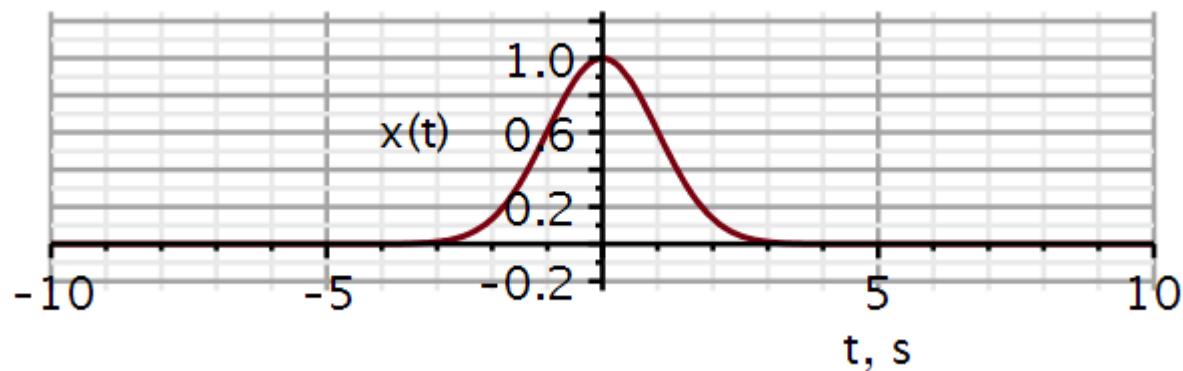
$$\begin{aligned} \int_{-\infty}^{\infty} x(at) e^{-j\omega t} dt &= \frac{1}{a} \int_{-\infty}^{\infty} x(s) e^{-j\frac{\omega}{a}s} ds \\ &= \frac{1}{a} X\left(\frac{\omega}{a}\right) \end{aligned}$$

$$a < 0$$

$$\begin{aligned} \int_{-\infty}^{\infty} x(at) e^{-j\omega t} dt &= \frac{1}{a} \int_{\infty}^{-\infty} x(s) e^{-j\frac{\omega}{a}s} ds \\ &= \frac{-1}{a} X\left(\frac{\omega}{a}\right) \end{aligned}$$

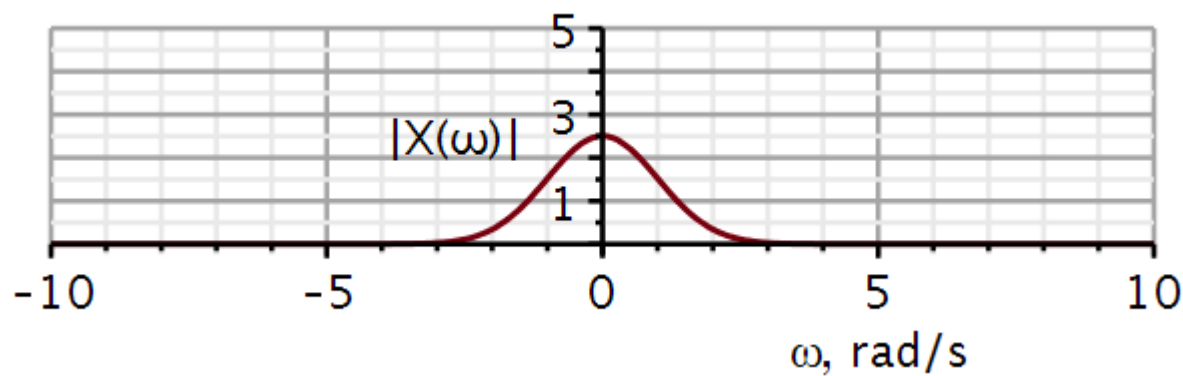
$$x(at) \leftrightarrow \frac{1}{|a|} X\left(\frac{\omega}{a}\right)$$

$$x(t) = e^{-t^2/2}$$

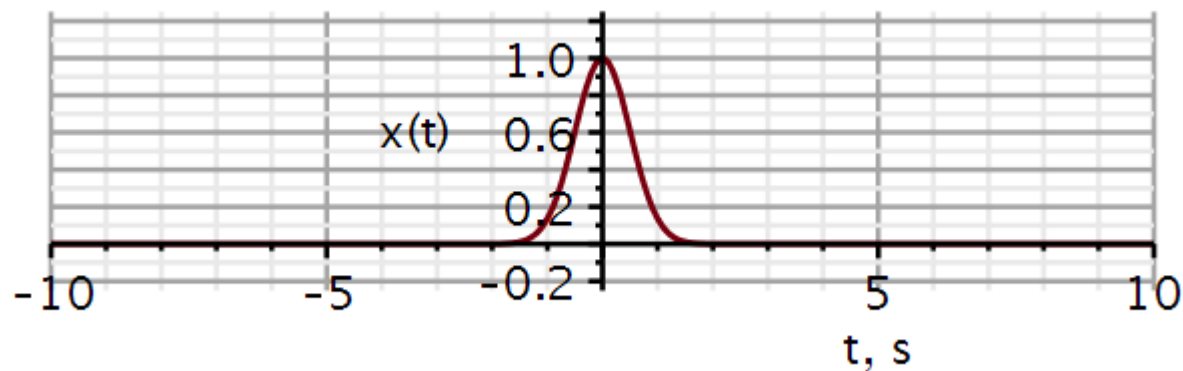


$$X1 := \text{fourier}(x1(t), t, \omega)$$

$$X1 := e^{-\frac{\omega^2}{2}} \sqrt{2} \sqrt{\pi}$$

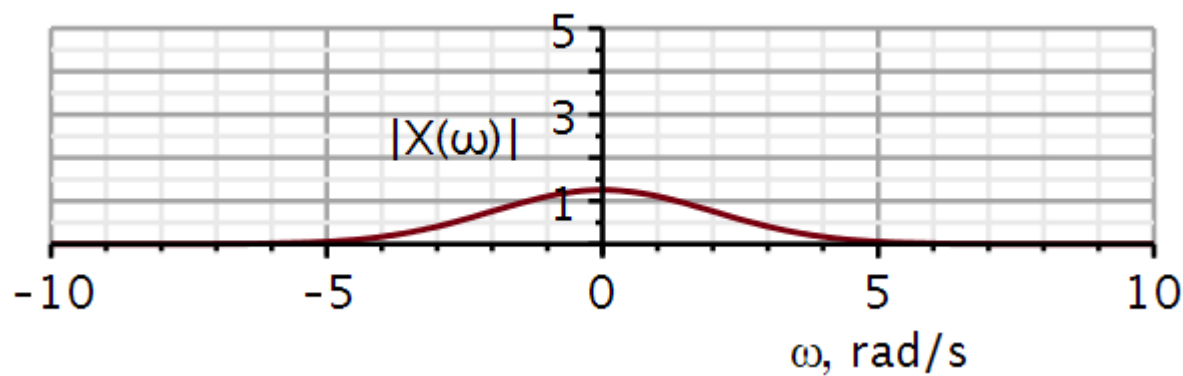


$$x(t) = e^{-(2t)^2/2}$$



$$X2 := \text{fourier}(x1(2 \cdot t), t, \omega)$$

$$X2 := \frac{e^{-\frac{\omega^2}{8}} \sqrt{2} \sqrt{\pi}}{2}$$



Using variable
substitution:

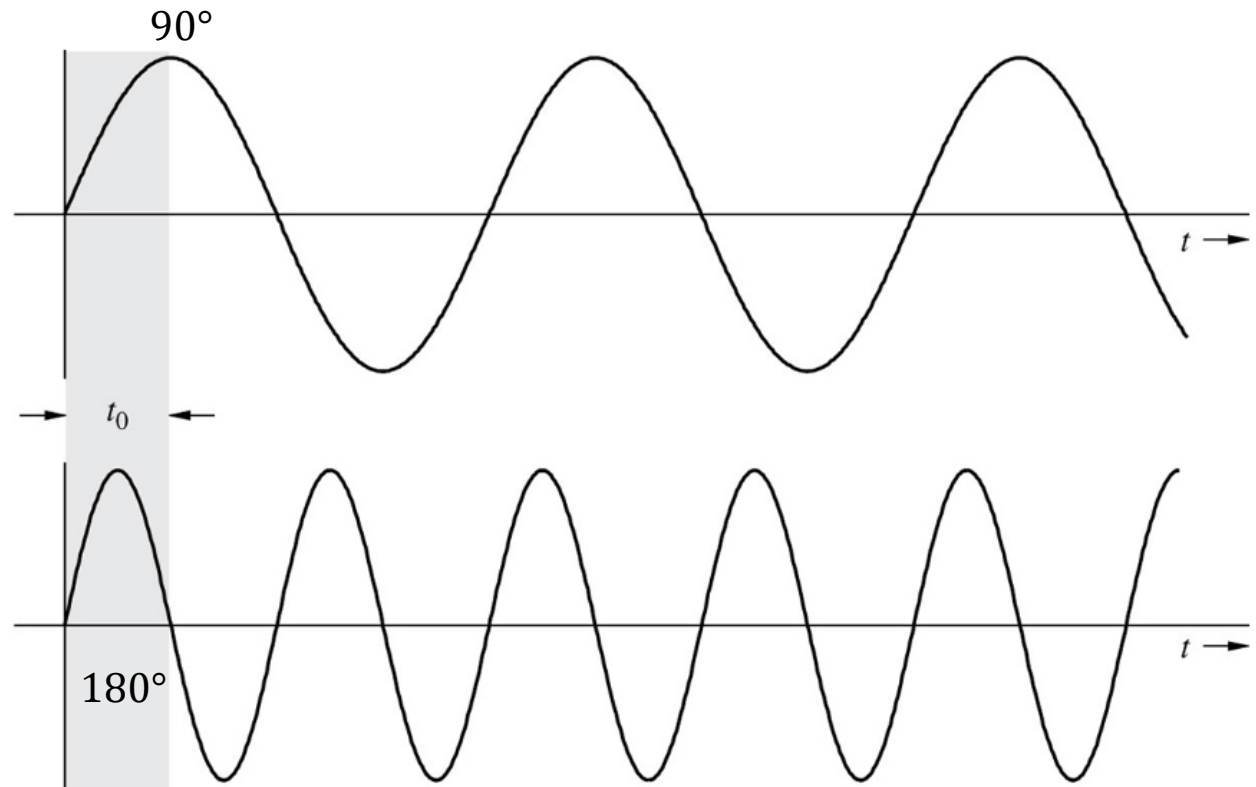
$$\begin{aligned}s &= t - t_0 \\ ds &= dt\end{aligned}$$

$$\begin{aligned}\int_{-\infty}^{\infty} x(t - t_0) e^{-j\omega t} dt &= \int_{-\infty}^{\infty} x(s) e^{-j\omega(s+t_0)} ds = e^{-j\omega t_0} \int_{-\infty}^{\infty} x(s) e^{-j\omega s} ds \\ &= e^{-j\omega t_0} X(\omega) = X(\omega) e^{j\theta}\end{aligned}$$

When a signal is delayed in the time domain, a phase angle is introduced in the frequency spectrum.

We notice that in the case of a time delay of the entire signal, the phase angle is a linear function of frequency:

$$\theta = -\omega \cdot t_0$$



Using variable substitution:

$$\begin{aligned}s &= t - t_0 \\ ds &= dt\end{aligned}$$

$$\int_{-\infty}^{\infty} x(t - t_0) e^{-j\omega t} dt = e^{-j\omega t_0} X(\omega)$$

Example:

$$\cos(\omega_0(t - t_d)) = \cos(\omega_0 t - \omega_0 t_d) = \cos(\omega_0 t + \theta)$$

$$t_d = \frac{-\theta}{\omega_0}$$

We set:

$$\begin{aligned}\theta &= -\pi/2 \\ t_d &= \frac{\pi}{2\omega_0}\end{aligned}$$

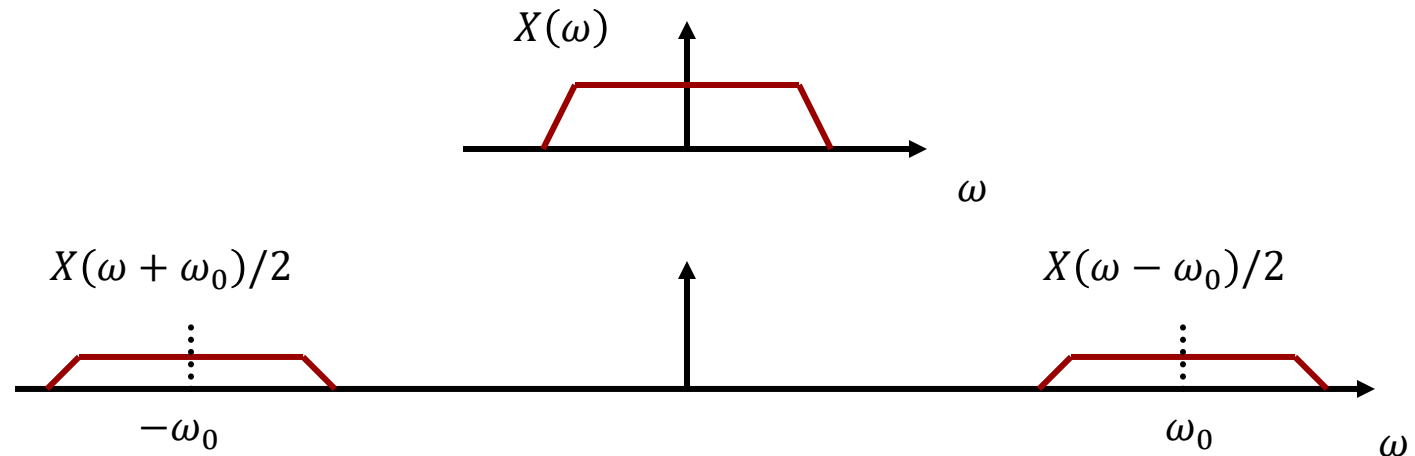
$$\begin{aligned}\cos(\omega_0(t - t_d)) &\leftrightarrow \pi[\delta(\omega - \omega_0) + \delta(\omega + \omega_0)] \cdot e^{-j\omega t_d} \\ &= \pi[\delta(\omega - \omega_0) \cdot e^{-j\omega_0 t_d} + \delta(\omega + \omega_0) \cdot e^{j\omega_0 t_d}] \\ &= \pi[\delta(\omega - \omega_0) \cdot e^{-j\pi/2} + \delta(\omega + \omega_0) \cdot e^{j\pi/2}] \\ \text{Why?} &= j\pi[\delta(\omega + \omega_0) - \delta(\omega - \omega_0)] \\ &\leftrightarrow \sin(\omega_0 t)\end{aligned}$$

Frequency shift:

$$x(t)e^{j\omega_0 t} \leftrightarrow \int_{-\infty}^{\infty} x(t)e^{j\omega_0 t} e^{-j\omega t} dt = \int_{-\infty}^{\infty} x(t)e^{-j(\omega - \omega_0)t} dt$$
$$= X(\omega - \omega_0)$$

Modulation:

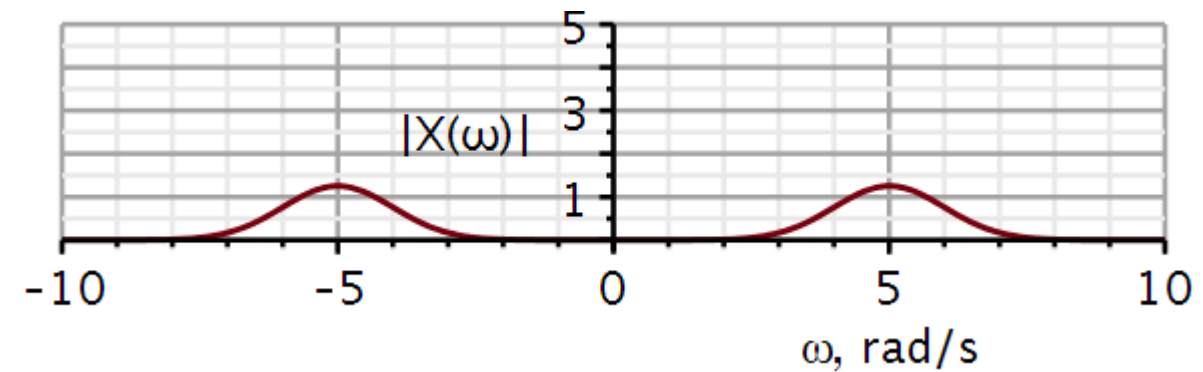
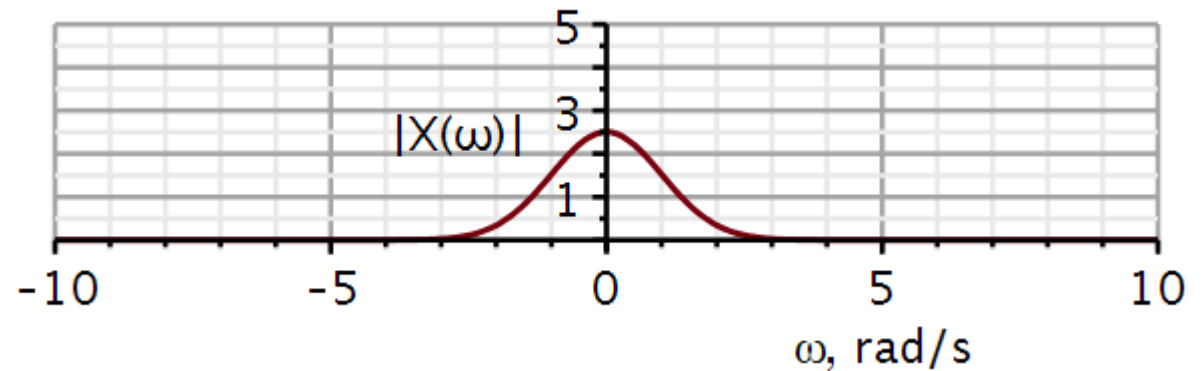
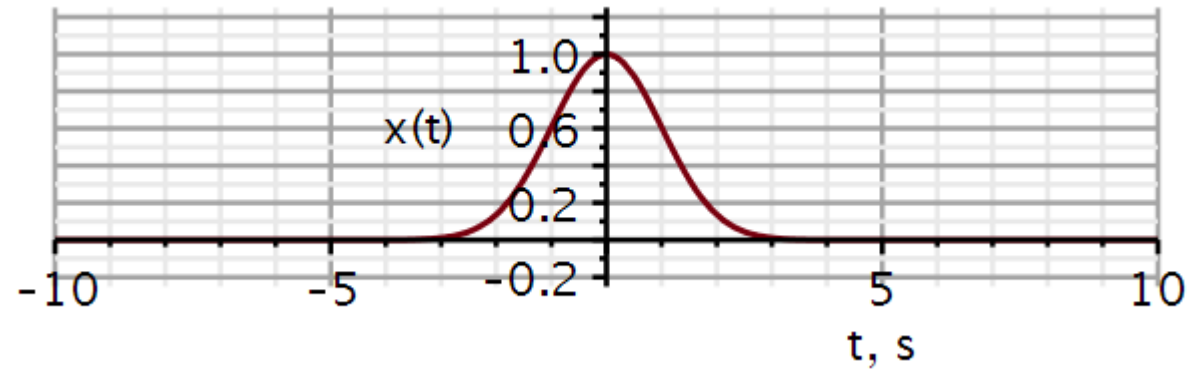
$$\begin{aligned}x(t) \cos(\omega_0 t) &= \frac{1}{2}x(t)e^{j\omega_0 t} + \frac{1}{2}x(t)e^{-j\omega_0 t} \\ &\Leftrightarrow \frac{1}{2}X(\omega - \omega_0) + \frac{1}{2}X(\omega + \omega_0)\end{aligned}$$



Here we modulate the Gauss function with a modulation frequency $\omega_0 = 5 \text{ rad/s}$.

$$X3 := \text{fourier}(x1(t) \cdot \cos(5 \cdot t), t, \omega)$$

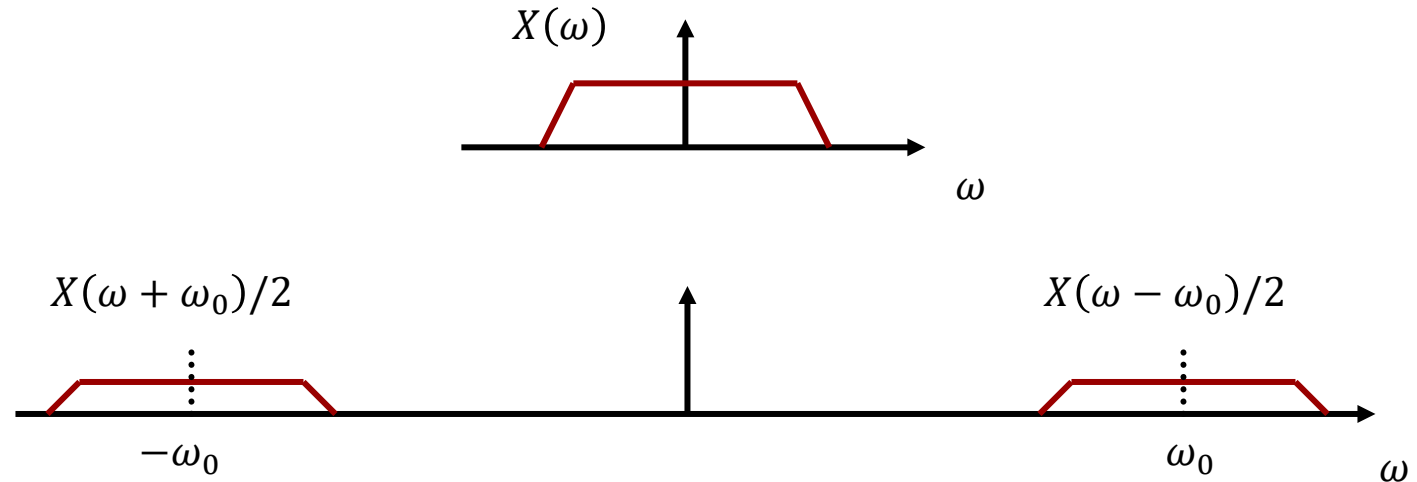
$$X3 := \sqrt{2} \sqrt{\pi} \cosh(5 \omega) e^{-\frac{25}{2} - \frac{\omega^2}{2}}$$



Modulation:

$$x(t) \cos(\omega_0 t) = \frac{1}{2} x(t) e^{j\omega_0 t} + \frac{1}{2} x(t) e^{-j\omega_0 t}$$

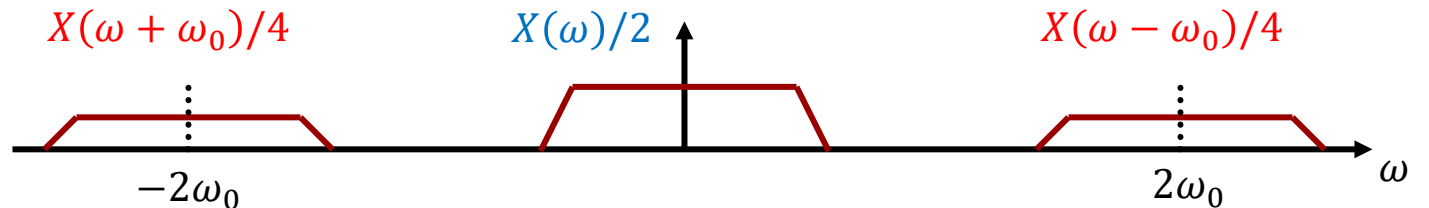
$$\Leftrightarrow \frac{1}{2} X(\omega - \omega_0) + \frac{1}{2} X(\omega + \omega_0)$$



Demodulation:

$$x(t) \cos(\omega_0 t) \cos(\omega_0 t) = \left(\frac{1}{2} x(t) e^{j\omega_0 t} + \frac{1}{2} x(t) e^{-j\omega_0 t} \right) \left(\frac{1}{2} e^{j\omega_0 t} + \frac{1}{2} e^{-j\omega_0 t} \right) = \left(\frac{1}{2} + \frac{1}{2} \cos(2\omega_0 t) \right) x(t)$$

$$\Leftrightarrow \frac{1}{4} X(\omega - 2\omega_0) + \frac{1}{4} X(\omega + 2\omega_0) + \frac{1}{2} X(\omega)$$



We find an extremely useful property by deriving the Fourier transform of the convolution integral:

Convolution of two functions in the time domain corresponds to multiplication of their spectra in the frequency domain

When you need to convolve two functions, it is often faster to Fourier transform the two functions, multiply their spectra, and then inverse Fourier transform the product.

The Fast Fourier Transformation (FFT) uses $n \times \log_2 n$ operations.

$$\begin{aligned}
 \int_{-\infty}^{\infty} x(\tau)h(t - \tau)d\tau &\leftrightarrow \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x(\tau)h(t - \tau)d\tau e^{-j\omega t}dt \\
 &= \int_{-\infty}^{\infty} x(\tau) \int_{-\infty}^{\infty} h(t - \tau)e^{-j\omega t}dt d\tau \\
 &= \int_{-\infty}^{\infty} x(\tau)H(\omega)e^{-j\omega \tau}d\tau \\
 &= H(\omega) \int_{-\infty}^{\infty} x(\tau)e^{-j\omega \tau}d\tau \\
 &= H(\omega) \underline{X(\omega)}
 \end{aligned}$$

$$\begin{array}{ccccc}
 y_{zs}(t) & = & h(t) & * & x(t) \\
 \uparrow_3 & & \downarrow_1 & & \downarrow_2 \\
 Y_{zs}(\omega) & = & H(\omega) & \cdot & X(\omega)
 \end{array}
 \quad n \times \log_2 n$$

$$1024 \times 10 \approx \underbrace{10^4}_{FFT} \ll \underbrace{10^6}_{convolution}$$

We also find an extremely useful property by deriving the inverse Fourier transform of the convolution integral:

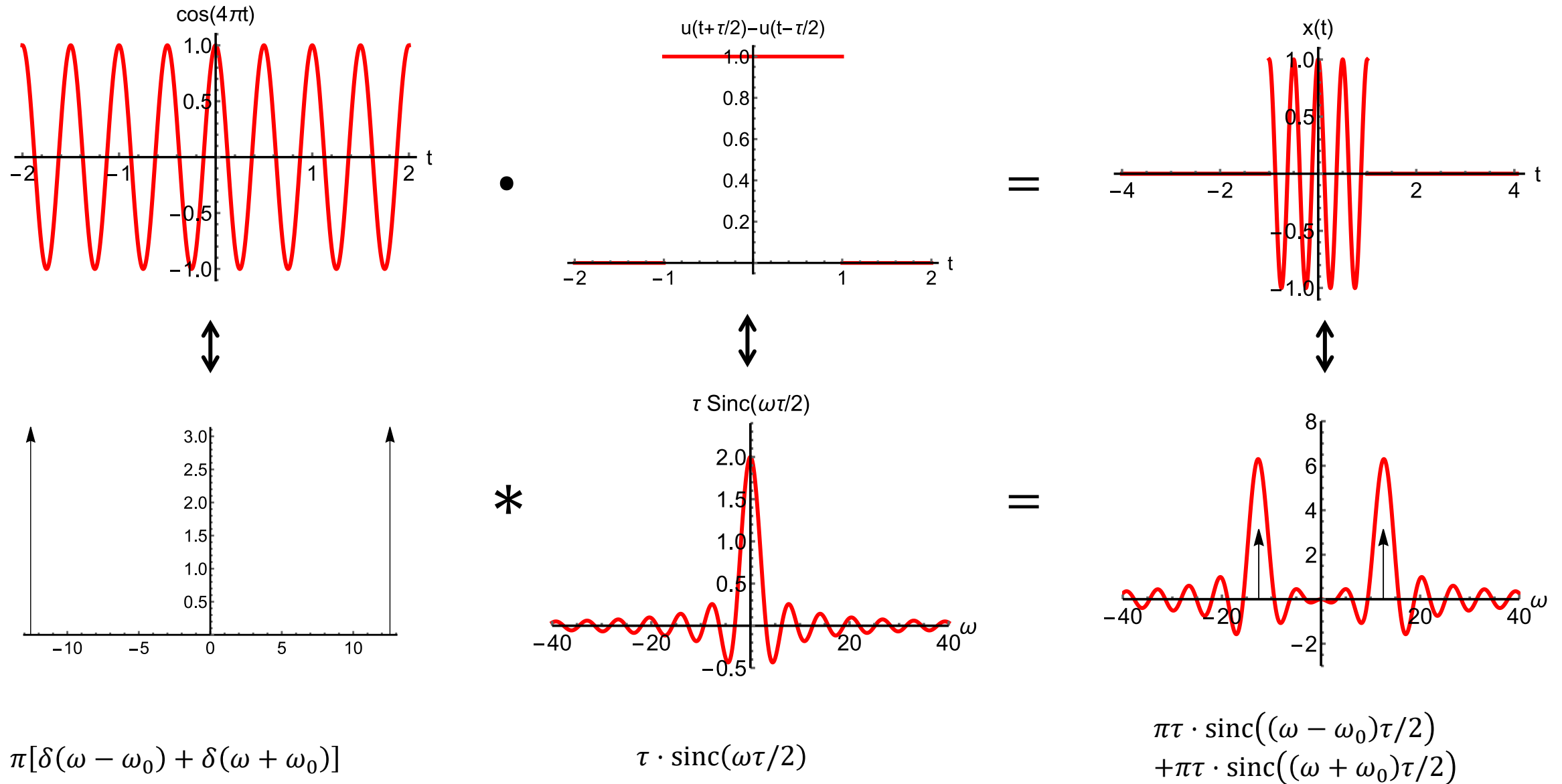
Convolution of two spectra in the frequency domain corresponds to multiplication of the functions in the time domain

When you need to convolve two spectra, it is often faster to inverse Fourier transform the two spectra, multiply their time functions, and then Fourier transform the product.

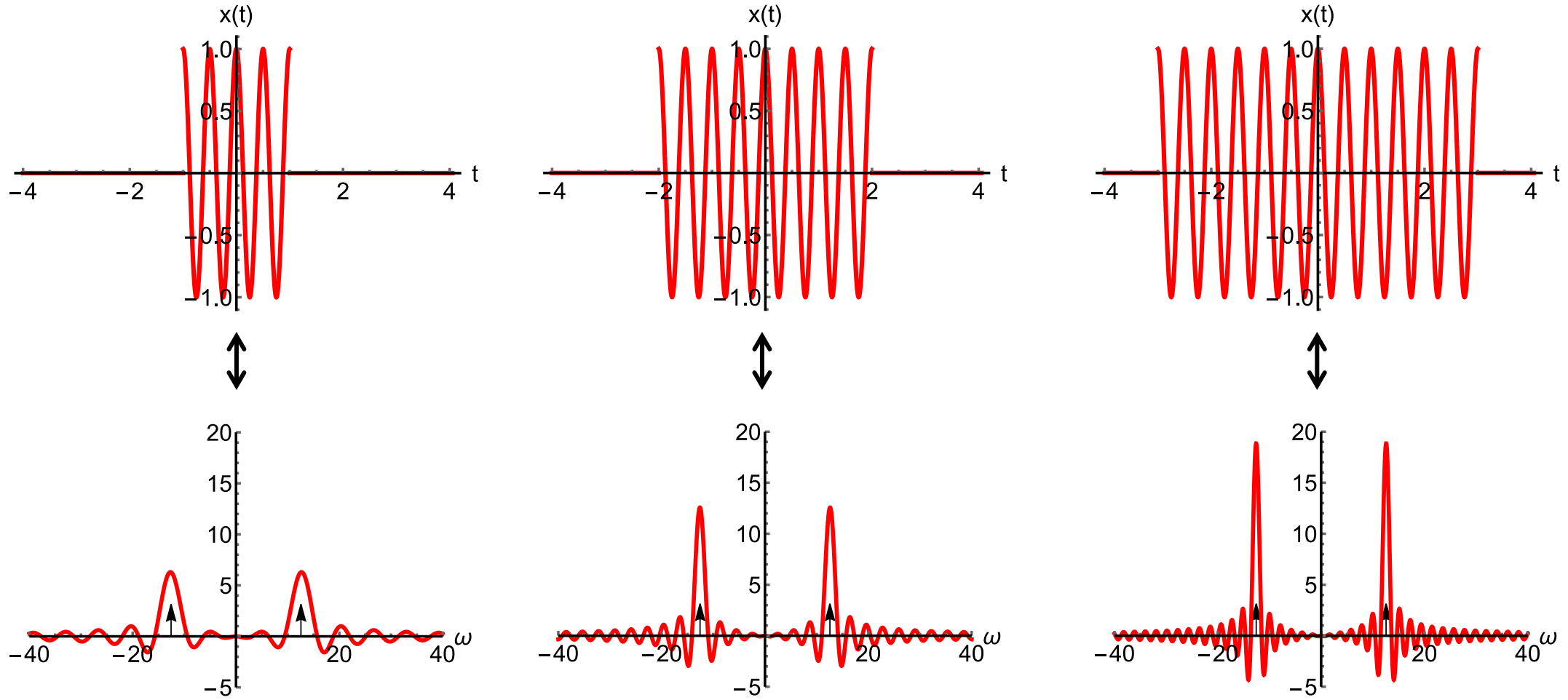
$$\begin{aligned}
 \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega_0) H(\omega - \omega_0) d\omega_0 &\leftrightarrow \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega_0) H(\omega - \omega_0) d\omega_0 e^{j\omega t} d\omega \\
 &= \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega_0) \frac{1}{2\pi} \int_{-\infty}^{\infty} H(\omega - \omega_0) e^{j\omega t} d\omega d\omega_0 \\
 &= \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega_0) h(t) e^{j\omega_0 t} d\omega_0 \\
 &= h(t) \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega_0) e^{j\omega_0 t} d\omega_0 \\
 &= h(t) x(t)
 \end{aligned}$$

$$\begin{array}{ccccc}
 y_{zs}(t) & = & 2\pi(h(t) & \cdot & x(t)) \\
 \downarrow_3 & & \uparrow_1 & & \uparrow_2 \\
 Y_{zs}(\omega) & = & H(\omega) & * & X(\omega)
 \end{array}$$

Convolution in the frequency domain



Convolution in the frequency domain



Leakage of energy

$$\int_{-\infty}^t x(\tau) d\tau = \int_{-\infty}^{\infty} x(\tau) u(t - \tau) d\tau = x(t) * u(t)$$

$$\Leftrightarrow X(\omega) \cdot \left[\pi \delta(\omega) + \frac{1}{j\omega} \right] = \pi X(0) \delta(\omega) + \frac{X(\omega)}{j\omega}$$

Example: Capacitor voltage

$$v_C(t) = \frac{1}{C} \int_{-\infty}^t i_C(\tau) d\tau$$

$$V_C(\omega) = \frac{\pi I_C(0) \delta(\omega)}{C} + \frac{I_C(\omega)}{j\omega C}$$

In case of zero DC, e.g. sinusoidal signal:

$$V_C(\omega) = \frac{I_C(\omega)}{j\omega C}$$

Impedance of capacitor:

$$Z_C(\omega) \stackrel{\text{def}}{=} \frac{V_C(\omega)}{I_C(\omega)} = \frac{1}{j\omega C} = \frac{1}{|\omega|C} e^{-j\pi/2}$$

$$\begin{aligned}\frac{dx}{dt} &= \frac{d}{dt} \left\{ \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{j\omega t} d\omega \right\} = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) \frac{d}{dt} e^{j\omega t} d\omega \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} [j\omega X(\omega)] e^{j\omega t} d\omega \leftrightarrow j\omega X(\omega)\end{aligned}$$

Example: Inductor voltage:

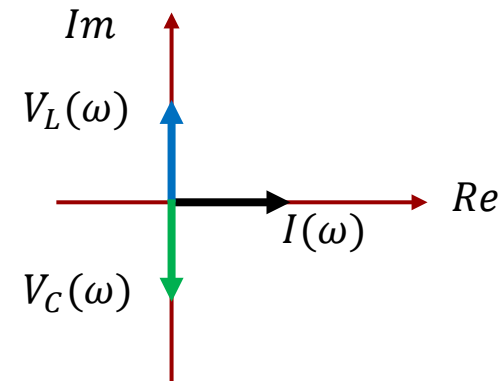
$$v_L(t) = L \frac{di_L}{dt} \leftrightarrow V_L(\omega) = j\omega L \cdot I_L(\omega)$$

Inductor impedance:

$$Z_L(\omega) = j\omega L = |\omega|L e^{j\pi/2}$$

The voltage drop across an inductor is 90 degrees ahead of the current. "Lead"

The voltage drop across a capacitor is 90 degrees behind the current. "Lag"



Application of the transform pair

$$\frac{d}{dt} \leftrightarrow j\omega$$

$$\ddot{y} + a_1\dot{y} + a_0y = b_2\ddot{x} + b_1\dot{x} + b_0x$$

$$F\{y(t)\} = Y(\omega) \quad F\{\dot{y}(t)\} = j\omega Y(\omega) \quad F\{\ddot{y}(t)\} = (j\omega)^2 Y(\omega)$$

$$[(j\omega)^2 + a_1 \cdot (j\omega) + a_0]Y(\omega) = [b_2 \cdot (j\omega)^2 + b_1 \cdot (j\omega) + b_0]X(\omega)$$

System frequency
characteristic:

$$\frac{Y(\omega)}{X(\omega)} = H(\omega) = \frac{b_2 \cdot (j\omega)^2 + b_1 \cdot (j\omega) + b_0}{(j\omega)^2 + a_1 \cdot (j\omega) + a_0} = \frac{P(j\omega)}{Q(j\omega)}$$

We now have the means to Fourier transform a differential equation in order to obtain the frequency characteristic of the system.

Conversely, if we have the frequency characteristic, we can translate it into the differential equation for the system. In many cases it might be faster to derive $H(\omega)$ from an electric circuit diagram first and then obtain the differential equation from $H(\omega)$.

In cases where we are only interested in $H(\omega)$, we do not need to derive the differential equation.

Lathi 4.4

Signal transmission in LTI systems

Video 2

Calculation of system response:

Time domain approach



Frequency domain approach



The impulse response $h(t)$ is of importance when we want to calculate system response in the time domain.

$$y(t) = h(t) * x(t) \quad \text{convolution}$$

If the purpose of the system is to attenuate certain frequencies, then the Fourier transform of the impulse response ($H(\omega)$) is a lot more useful.

$$Y(\omega) = X(\omega) \cdot H(\omega) = |X(\omega)| \cdot |H(\omega)| e^{j[\angle X(\omega) + \angle H(\omega)]}$$

Its modulus is the gain of the system as a function of frequency.

$$|Y(\omega)| = |X(\omega)| \cdot |H(\omega)|$$

Amplitude response:

Its angle reveals the time delay as a function of frequency.

$$\angle Y(\omega) = \angle X(\omega) + \angle H(\omega)$$

Phase response:

A distortion-less system is one that only scales and delays the signal as it passes through the system:

Here $y(t)$ is a scaled and delayed copy of $x(t)$. Apart from these changes, the output is identical to the input, and the system is distortion free.

Time domain approach

$$x(t) \longrightarrow \boxed{* h(t)} \longrightarrow y(t)$$

Frequency domain approach

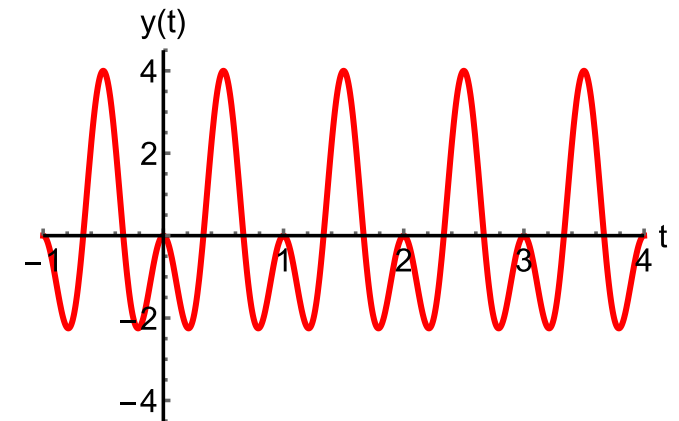
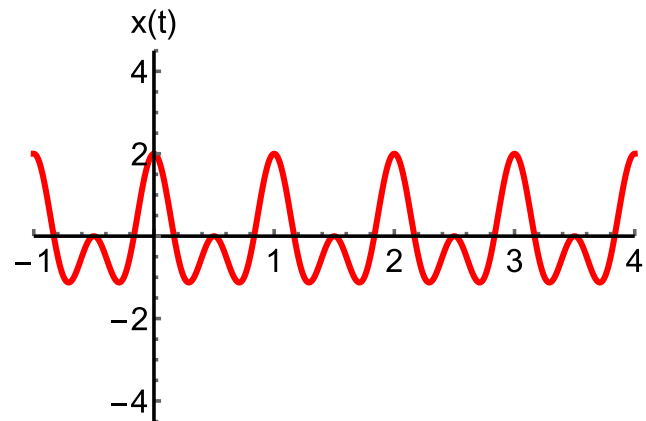
$$X(\omega) \longrightarrow \boxed{\cdot H(\omega)} \longrightarrow Y(\omega)$$

$$y(t) = k \cdot x(t - t_d)$$

F. ex.

$$x(t) = \cos(\omega_1 t) + \cos(\omega_2 t)$$

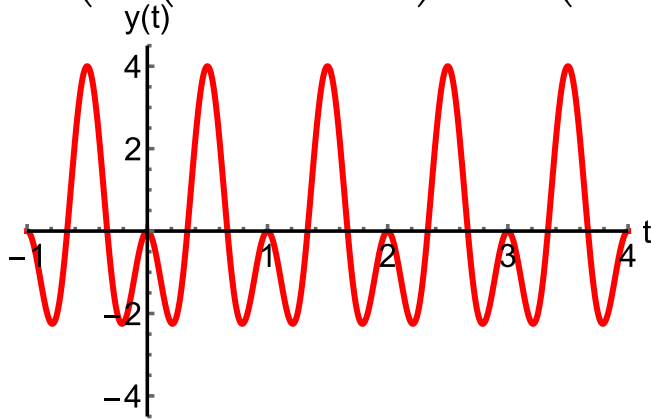
$$y(t) = k \cdot (\cos(\omega_1(t - t_d)) + \cos(\omega_2(t - t_d)))$$



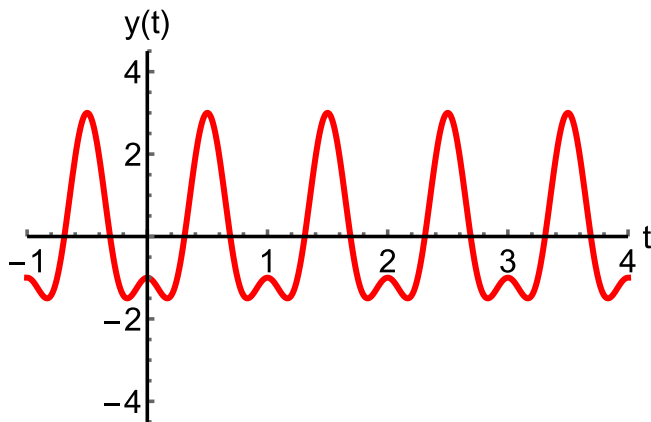
Amplitude and phase distortion

Amplitude distortion

$$y(t) = 2 \cdot \left(\cos \left(2\pi \cdot 1 \cdot \left(t - \frac{1}{2} \right) \right) + \cos \left(2\pi \cdot 2 \cdot \left(t - \frac{1}{2} \right) \right) \right)$$

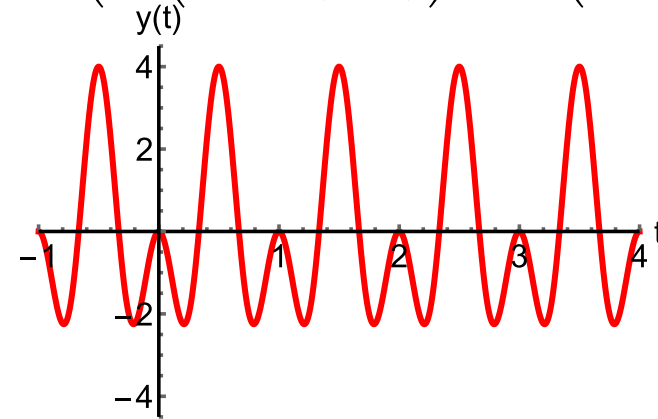


$$y(t) = 2 \cos \left(2\pi \cdot 1 \cdot \left(t - \frac{1}{2} \right) \right) + \cos \left(2\pi \cdot 2 \cdot \left(t - \frac{1}{2} \right) \right)$$

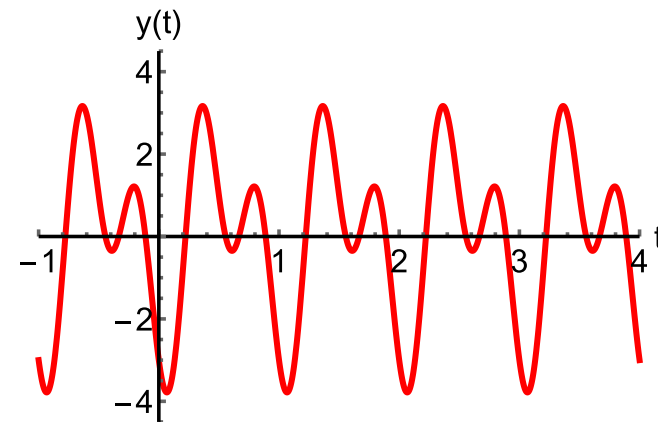


Phase distortion

$$y(t) = 2 \cdot \left(\cos \left(2\pi \cdot 1 \cdot \left(t - \frac{1}{2} \right) \right) + \cos \left(2\pi \cdot 2 \cdot \left(t - \frac{1}{2} \right) \right) \right)$$



$$y(t) = 2 \cdot \left(\cos \left(2\pi \cdot 1 \cdot \left(t - \frac{1}{2} \right) \right) + \cos \left(2\pi \cdot 2 \cdot \left(t - \frac{2}{3} \right) \right) \right)$$



A system with no amplitude distortion will have an amplitude characteristic:

$$|H(\omega)| = k$$

We also need to understand what is required of a system to avoid phase distortion, i.e., to delay all frequencies by the same constant time delay.

$$y(t) = k \cdot (\cos(\omega_1(t - t_d)) + \cos(\omega_2(t - t_d)))$$

We introduce the phase angle into the expression for the signal and match it with the time delay.

$$y(t) = k \cdot (\cos(\omega_1 t + \theta(\omega_1)) + \cos(\omega_2 t + \theta(\omega_2)))$$

We observe that the time delay is the negative derivative of the phase angle with respect to angular frequency.

$$\theta(\omega) = -\omega t_d \Rightarrow \frac{d\theta(\omega)}{d\omega} = -t_d$$

To have constant time delay for all frequencies, the phase angle must be a linear function of frequency. The system is said to have a **linear phase characteristic**.

$$t_d = -\frac{d\theta(\omega)}{d\omega}$$

$$t_d = -\frac{d\theta(\omega)}{d\omega} = \text{constant} \Rightarrow \theta(\omega) = \underbrace{\text{constant} \cdot \omega}_{\text{linear}}$$

Ideal filter: $H(\omega) = \text{rect}\left(\frac{\omega}{2W}\right) \cdot e^{-j\omega t_d}$

The ideal filter has a gain of 1 in the pass band, a gain of 0 in the stop band, and a linear phase.

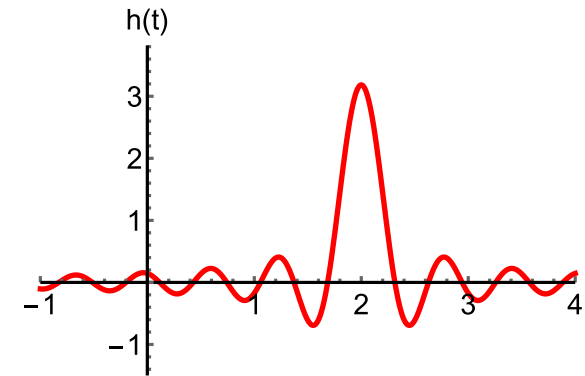
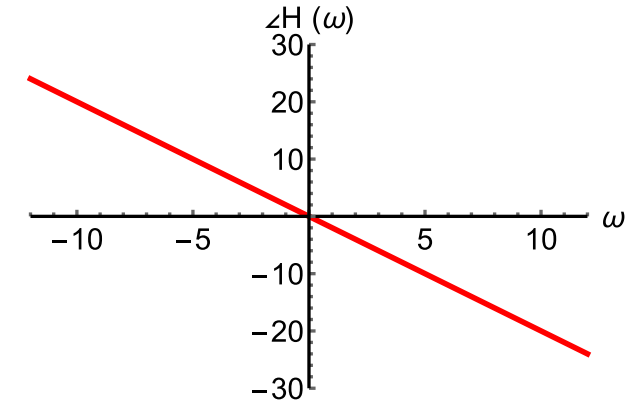
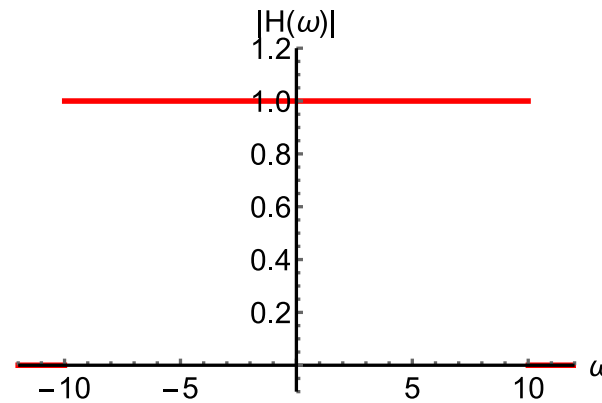
Non-causal filter is unrealizable:

$$h(t) = \frac{W}{\pi} \text{sinc}(W(t - t_d))$$

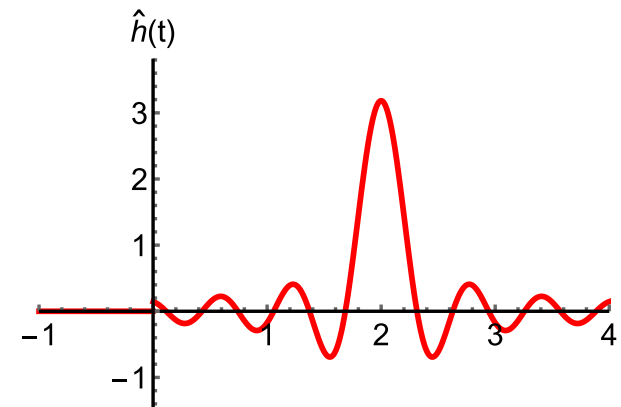
Practical filter:

Causal filter is realizable:

$$\hat{h}(t) = \frac{W}{\pi} \text{sinc}(W(t - t_d)) \cdot u(t)$$



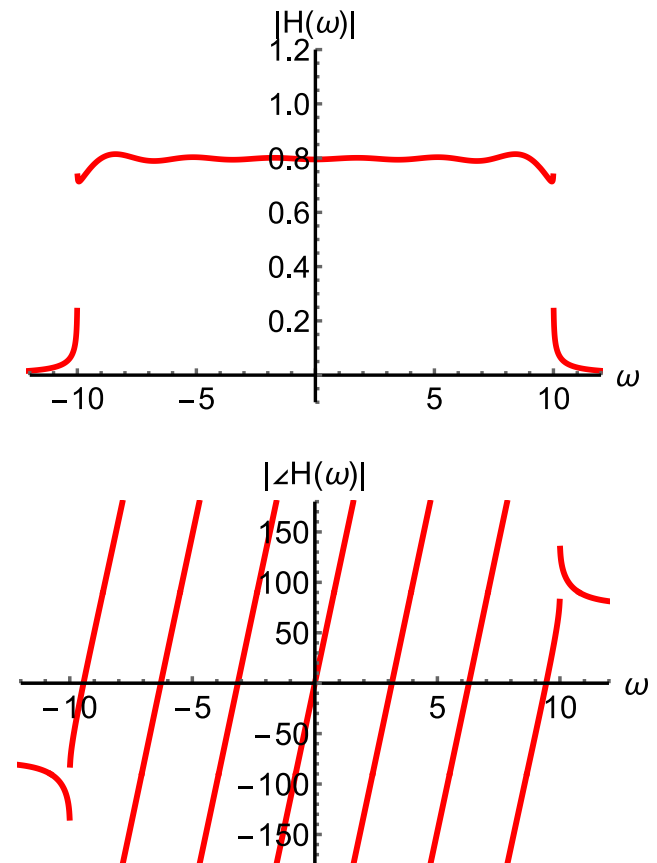
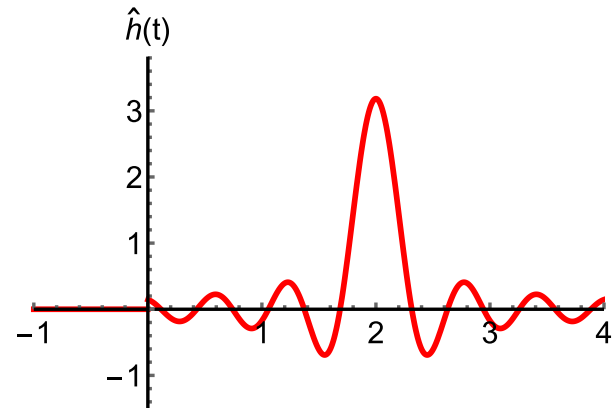
Here the impulse response starts before the impulse.



$$\hat{h}(t) = \frac{W}{\pi} \text{sinc}(W(t - t_d)) \cdot u(t)$$

The consequence of truncating the impulse response at $t = 0$ is shown here.

We get ripple on the amplitude characteristic and the phase characteristic is not linear.



Problems

Problem 1

Derive these results using pen and paper:

$$F\{\delta(t)\} = \int_{-\infty}^{\infty} \delta(t)e^{-j\omega t} dt = ?$$

Answer

1

$$F^{-1}\{\delta(\omega)\} = \frac{1}{2\pi} \int_{-\infty}^{\infty} \delta(\omega)e^{j\omega t} d\omega = ?$$

$\frac{1}{2\pi}$

$$F^{-1}\{2\pi\delta(\omega)\} = \int_{-\infty}^{\infty} \delta(\omega)e^{j\omega t} d\omega = ?$$

1

$$F^{-1}\{2\pi\delta(\omega - \omega_0)\} = \int_{-\infty}^{\infty} \delta(\omega - \omega_0)e^{j\omega t} d\omega = ?$$

$e^{j\omega_0 t}$

$$F^{-1}\{\pi(\delta(\omega - \omega_0) + \delta(\omega + \omega_0))\} = ?$$

$\frac{e^{j\omega_0 t} + e^{-j\omega_0 t}}{2} = \cos(\omega_0 t)$

Problem 2

Derive these results using Maple/Python:

$$F\{\delta(t)\} = \int_{-\infty}^{\infty} \delta(t)e^{-j\omega t} dt = ?$$

Answer

1

$$F^{-1}\{\delta(\omega)\} = \frac{1}{2\pi} \int_{-\infty}^{\infty} \delta(\omega)e^{j\omega t} d\omega = ?$$

$\frac{1}{2\pi}$

$$F^{-1}\{2\pi\delta(\omega)\} = \int_{-\infty}^{\infty} \delta(\omega)e^{j\omega t} d\omega = ?$$

1

$$F^{-1}\{2\pi\delta(\omega - \omega_0)\} = \int_{-\infty}^{\infty} \delta(\omega - \omega_0)e^{j\omega t} d\omega = ?$$

$e^{j\omega_0 t}$

$$F^{-1}\{\pi(\delta(\omega - \omega_0) + \delta(\omega + \omega_0))\} = ?$$

$\frac{e^{j\omega_0 t} + e^{-j\omega_0 t}}{2} = \cos(\omega_0 t)$

Problem 3

Observe the two functions plotted here.

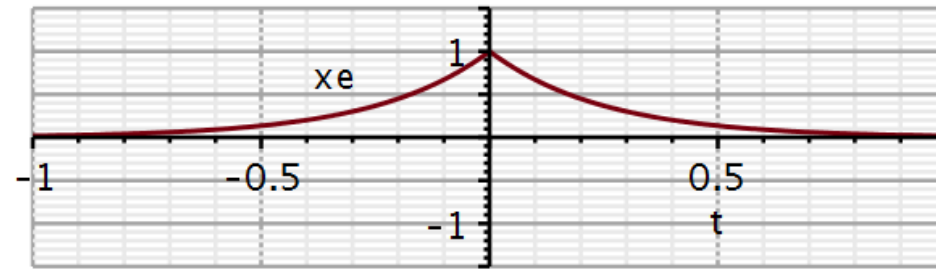
x_e is even and x_o is odd.

1. What symmetries do you expect in the Fourier transforms?
2. Calculate their Fourier transforms and verify your expectations.

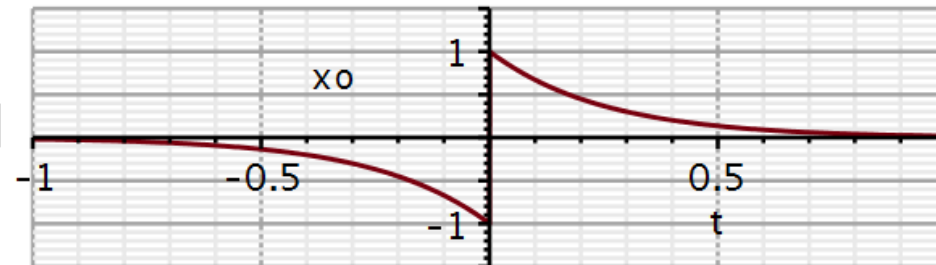
$$x_e := e^{-4 \cdot |t|} :$$

$$x_o := -e^{-4 \cdot |t|} \cdot \text{Heaviside}(-t) + e^{-4 \cdot |t|} \cdot \text{Heaviside}(t) :$$

```
plot(xe, t=-1 ..1, -1.5 ..1.5, thickness = 3, axesfont = [Helvetica,
    "roman", 18], axis[2] = [thickness = 2.5], axis[1] = [thickness
    = 2.5], labels = ["t", "xe"], labelfont = [Helvetica, 18], numpoints
    = 100, gridlines, size = [600, 200])
```



```
plot(xo, t=-1 ..1, -1.5 ..1.5, thickness = 3, axesfont = [Helvetica,
    "roman", 18], axis[2] = [thickness = 2.5], axis[1] = [thickness
    = 2.5], labels = ["t", "xo"], labelfont = [Helvetica, 18], numpoints
    = 100, gridlines, size = [600, 200])
```



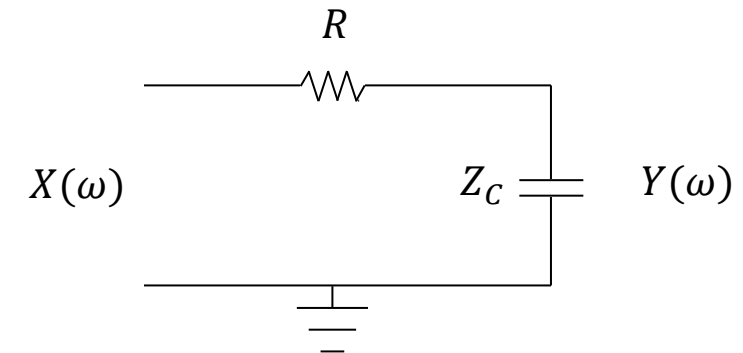
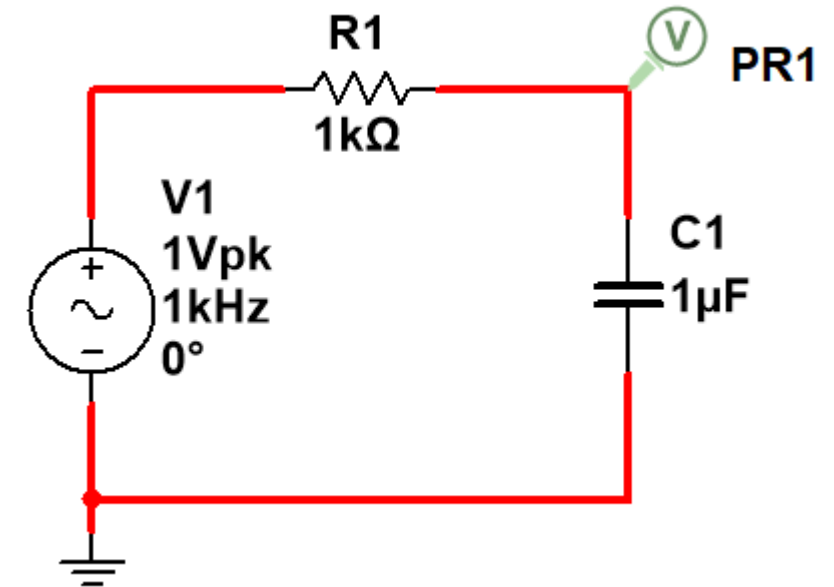
Problem 4. RC circuit analysis in frequency domain

1. Sketch this RC circuit in KiCad/Spice.
2. Run an AC sweep in the frequency range from 1 Hz to 10 kHz and plot magnitude in decibel and phase in degrees.
3. With pen and paper, redraw the circuit with the capacitor replaced by its impedance.
4. Show that the input-output relationship can be expressed as a voltage divider:

$$Y(\omega) = \frac{Z_C(\omega)}{R + Z_C(\omega)} X(\omega)$$

5. Insert the expression for the impedance of a capacitor and show that:

$$H(\omega) = \frac{Y(\omega)}{X(\omega)} = \frac{1}{j\omega RC + 1} = \frac{\frac{1}{RC}}{j\omega + \frac{1}{RC}}$$



Problem 4. RC circuit analysis in frequency domain

6. With pen and paper, determine the symmetry properties of the amplitude characteristic and the phase characteristic.
7. Plot the magnitude and phase plots in Maple/Python
8. Notice how different the curves in KiCad/Spice looks. Always take notice of the type of axes used (linear, log, decibel).
9. Use semilogplot to plot the spectra on logarithmic frequency axes in Hz. In this case, plot only positive frequencies. Plot amplitude in decibels.
10. Calculate the inverse Fourier transform of $H(\omega)$ and plot it.
11. Replace the 1k resistor with a 500 Ohm resistor. Repeat questions 7 and 10.
12. Compare the two cases and explain using the time scaling property of Fourier transformation.
13. Plot the time delay as a function of frequency in Maple/Python

Problem solutions

Problem 1

Derive these results using pen and paper:

$$F\{\delta(t)\} = \int_{-\infty}^{\infty} \delta(t)e^{-j\omega t} dt = ?$$

Answer

1

$$F^{-1}\{\delta(\omega)\} = \frac{1}{2\pi} \int_{-\infty}^{\infty} \delta(\omega)e^{j\omega t} d\omega = ?$$

$\frac{1}{2\pi}$

$$F^{-1}\{2\pi\delta(\omega)\} = \int_{-\infty}^{\infty} \delta(\omega)e^{j\omega t} d\omega = ?$$

1

$$F^{-1}\{2\pi\delta(\omega - \omega_0)\} = \int_{-\infty}^{\infty} \delta(\omega - \omega_0)e^{j\omega t} d\omega = ?$$

$e^{j\omega_0 t}$

$$F^{-1}\{\pi(\delta(\omega - \omega_0) + \delta(\omega + \omega_0))\} = ?$$

$\frac{e^{j\omega_0 t} + e^{-j\omega_0 t}}{2} = \cos(\omega_0 t)$

Problem 1 (sol)

Derive these results using pen and paper:

$$F\{\delta(t)\} = \int_{-\infty}^{\infty} \delta(t)e^{-j\omega t} dt = \int_{-\infty}^{\infty} \delta(t)e^{-j\omega_0 t} dt = 1$$

$$F^{-1}\{\delta(\omega)\} = \frac{1}{2\pi} \int_{-\infty}^{\infty} \delta(\omega)e^{j\omega t} d\omega = \frac{1}{2\pi} \int_{-\infty}^{\infty} \delta(\omega)e^{j\omega_0 t} d\omega = \frac{1}{2\pi}$$

$$F^{-1}\{2\pi\delta(\omega)\} = \int_{-\infty}^{\infty} \delta(\omega)e^{j\omega t} d\omega = \int_{-\infty}^{\infty} \delta(\omega)e^{j\omega_0 t} d\omega = 1$$

$$F^{-1}\{2\pi\delta(\omega - \omega_0)\} = \int_{-\infty}^{\infty} \delta(\omega - \omega_0)e^{j\omega t} d\omega = \int_{-\infty}^{\infty} \delta(\omega - \omega_0)e^{j\omega_0 t} d\omega = e^{j\omega_0 t}$$

$$F^{-1}\{\pi(\delta(\omega - \omega_0) + \delta(\omega + \omega_0))\} = \frac{e^{j\omega_0 t} + e^{-j\omega_0 t}}{2} = \cos(\omega_0 t)$$

Problem 2

Derive these results using Maple:

$$F\{\delta(t)\} = \int_{-\infty}^{\infty} \delta(t)e^{-j\omega t} dt = ?$$

Answer

1

$$F^{-1}\{\delta(\omega)\} = \frac{1}{2\pi} \int_{-\infty}^{\infty} \delta(\omega)e^{j\omega t} d\omega = ?$$

$\frac{1}{2\pi}$

$$F^{-1}\{2\pi\delta(\omega)\} = \int_{-\infty}^{\infty} \delta(\omega)e^{j\omega t} d\omega = ?$$

1

$$F^{-1}\{2\pi\delta(\omega - \omega_0)\} = \int_{-\infty}^{\infty} \delta(\omega - \omega_0)e^{j\omega t} d\omega = ?$$

$e^{j\omega_0 t}$

$$F^{-1}\{\pi(\delta(\omega - \omega_0) + \delta(\omega + \omega_0))\} = ?$$

$\frac{e^{j\omega_0 t} + e^{-j\omega_0 t}}{2} = \cos(\omega_0 t)$

Problem 2 (sol)

Derive these results using Maple:

$$F\{\delta(t)\} = \int_{-\infty}^{\infty} \delta(t) e^{-j\omega t} dt = ?$$

$$F^{-1}\{\delta(\omega)\} = \frac{1}{2\pi} \int_{-\infty}^{\infty} \delta(\omega) e^{j\omega t} d\omega = ?$$

$$F^{-1}\{2\pi\delta(\omega)\} = \int_{-\infty}^{\infty} \delta(\omega) e^{j\omega t} d\omega = ?$$

$$F^{-1}\{2\pi\delta(\omega - \omega_0)\} = \int_{-\infty}^{\infty} \delta(\omega - \omega_0) e^{j\omega t} d\omega = ?$$

$$F^{-1}\{\pi(\delta(\omega - \omega_0) + \delta(\omega + \omega_0))\} = ?$$

Answer

→ $x := (\text{Heaviside}(t + 0.5) - \text{Heaviside}(t - 0.5)) :$
 $\delta := t \rightarrow \text{Dirac}(t) :$
 $\text{sgn} := t \rightarrow 2 \cdot \text{Heaviside}(t) - 1 :$

fourier(Dirac(t), t, ω)

1

invfourier(δ(ω), ω, t)

$\frac{1}{2\pi}$

invfourier(2 π · δ(ω), ω, t)

1

invfourier(2 π · δ(ω - ω₀), ω, t)

$e^{j\omega_0 t}$

invfourier(π · (δ(ω - ω₀) + δ(ω + ω₀)), ω, t)

$\cos(\omega_0 t)$

Problem 3

Observe the two functions plotted here.

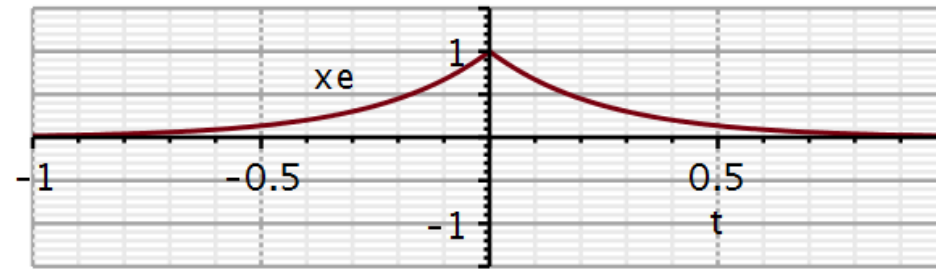
x_e is even and x_o is odd.

1. What symmetries do you expect in the Fourier transforms?
2. Calculate their Fourier transforms and verify your expectations.

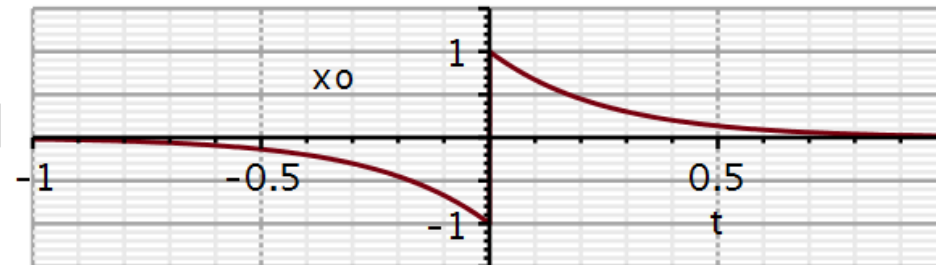
$$x_e := e^{-4 \cdot |t|} :$$

$$x_o := -e^{-4 \cdot |t|} \cdot \text{Heaviside}(-t) + e^{-4 \cdot |t|} \cdot \text{Heaviside}(t) :$$

```
plot(xe, t=-1 ..1, -1.5 ..1.5, thickness = 3, axesfont = [Helvetica,
    "roman", 18], axis[2] = [thickness = 2.5], axis[1] = [thickness
    = 2.5], labels = ["t", "xe"], labelfont = [Helvetica, 18], numpoints
    = 100, gridlines, size = [600, 200])
```



```
plot(xo, t=-1 ..1, -1.5 ..1.5, thickness = 3, axesfont = [Helvetica,
    "roman", 18], axis[2] = [thickness = 2.5], axis[1] = [thickness
    = 2.5], labels = ["t", "xo"], labelfont = [Helvetica, 18], numpoints
    = 100, gridlines, size = [600, 200])
```



Problem 3 (sol)

Theory predicts real and even functions to have a Fourier transform that is **real and even** in ω .

Theory also predicts real and odd functions to have a Fourier transform that is **imaginary and odd** in ω .

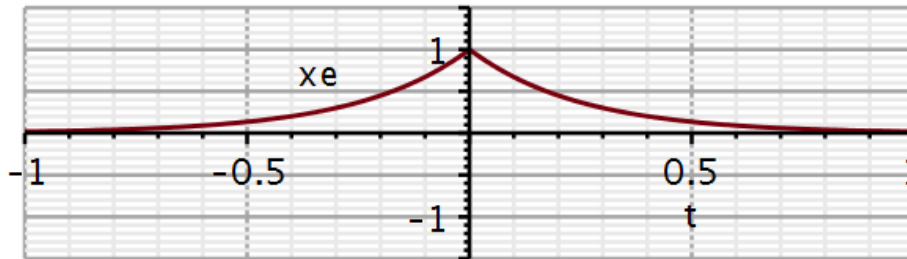
$$\begin{aligned}
 X(\omega) &= \int_{-\infty}^{\infty} (x_e(t) + x_o(t))(\cos \omega t + j \sin \omega t) dt \\
 &= \underbrace{\int_{-\infty}^{\infty} (x_e(t))(\cos \omega t) dt}_{\text{real part is even in } \omega} + \underbrace{\int_{-\infty}^{\infty} (x_e(t))(j \sin \omega t) dt}_{=0} \\
 &\quad + \underbrace{\int_{-\infty}^{\infty} (x_o(t))(\cos \omega t) dt}_{=0} + j \underbrace{\int_{-\infty}^{\infty} (x_o(t))(\sin \omega t) dt}_{\text{imaginary part is odd in } \omega}
 \end{aligned}$$

$$\left. \begin{aligned} Re\{X(\omega)\} &= Re\{X(-\omega)\} \\ Im\{X(\omega)\} &= -Im\{X(-\omega)\} \end{aligned} \right\} \Rightarrow \begin{cases} |X(\omega)| = |X(-\omega)| \\ \angle X(\omega) = -\angle X(-\omega) \end{cases} \quad \begin{array}{l} \text{Even function} \\ \text{Odd function} \end{array}$$

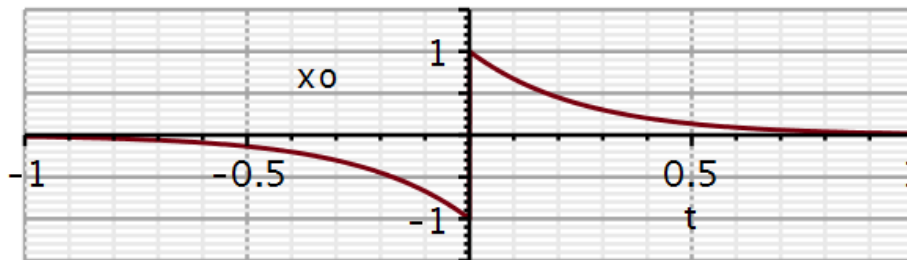
Problem 3 (sol)

```
xe := e-4·|t| ;
xo := -e-4·|t|·Heaviside(-t) + e-4·|t|·Heaviside(t) :
```

```
plot(xe, t=-1..1, -1.5..1.5, thickness=3, axesfont=[Helvetica,
"roman", 18], axis[2]=[thickness=2.5], axis[1]=[thickness
=2.5], labels=["t", "xe"], labelfont=[Helvetica, 18], numpoints
=100, gridlines, size=[600, 200])
```



```
plot(xo, t=-1..1, -1.5..1.5, thickness=3, axesfont=[Helvetica,
"roman", 18], axis[2]=[thickness=2.5], axis[1]=[thickness
=2.5], labels=["t", "xo"], labelfont=[Helvetica, 18], numpoints
=100, gridlines, size=[600, 200])
```



Calculating Fourier transform

$X := \text{fourier}(xe, t, \omega)$

$$X := \frac{8}{\omega^2 + 16}$$

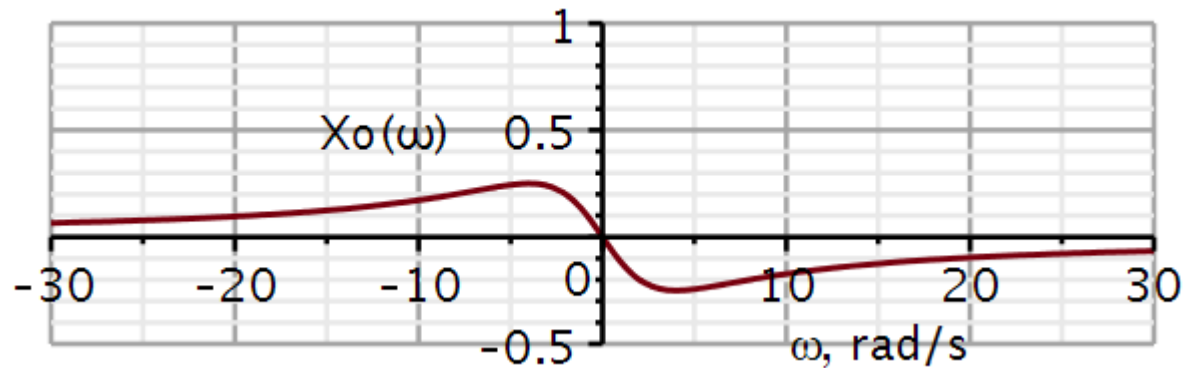
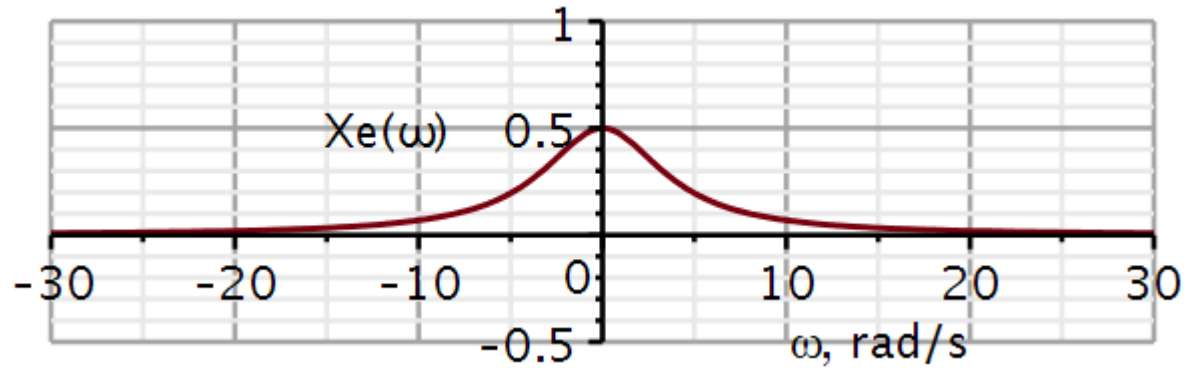
$X := \text{fourier}(xo, t, \omega)$

$$X := \frac{-2I\omega}{\omega^2 + 16}$$

For an even function, the *FT* is real and even.

For an odd function, the *FT* is imaginary and odd.

Problem 3 (sol)



Calculating Fourier transform

$$X := \text{fourier}(xe, t, \omega)$$

$$X := \frac{8}{\omega^2 + 16}$$

$$X := \text{fourier}(xo, t, \omega)$$

$$X := \frac{-2I\omega}{\omega^2 + 16}$$

For an even function, the *FT* is real and even.

For an odd function, the *FT* is imaginary and odd.

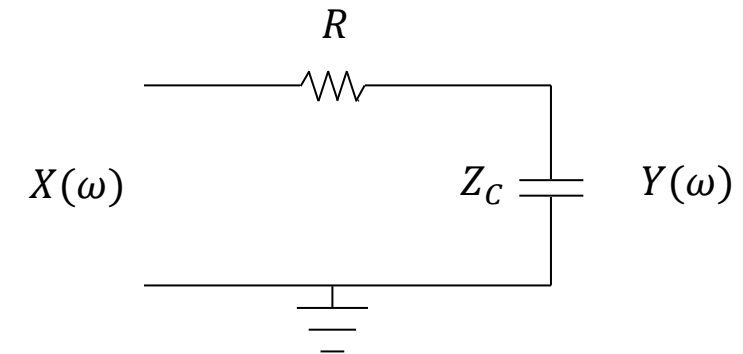
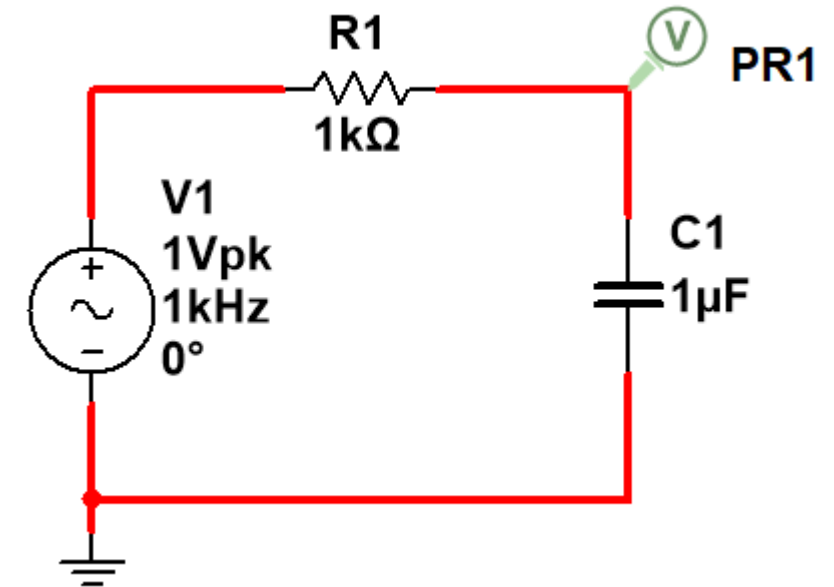
Problem 4. RC circuit analysis in frequency domain

1. Sketch this RC circuit in KiCad/Spice.
2. Run an AC sweep in the frequency range from 1 Hz to 10 kHz and plot magnitude in decibel and phase in degrees.
3. With pen and paper, redraw the circuit with the capacitor replaced by its impedance.
4. Show that the input-output relationship can be expressed as a voltage divider:

$$Y(\omega) = \frac{Z_C(\omega)}{R + Z_C(\omega)} X(\omega)$$

5. Insert the expression for the impedance of a capacitor and show that:

$$H(\omega) = \frac{Y(\omega)}{X(\omega)} = \frac{1}{j\omega RC + 1} = \frac{\frac{1}{RC}}{j\omega + \frac{1}{RC}}$$



Problem 4. RC circuit analysis in frequency domain

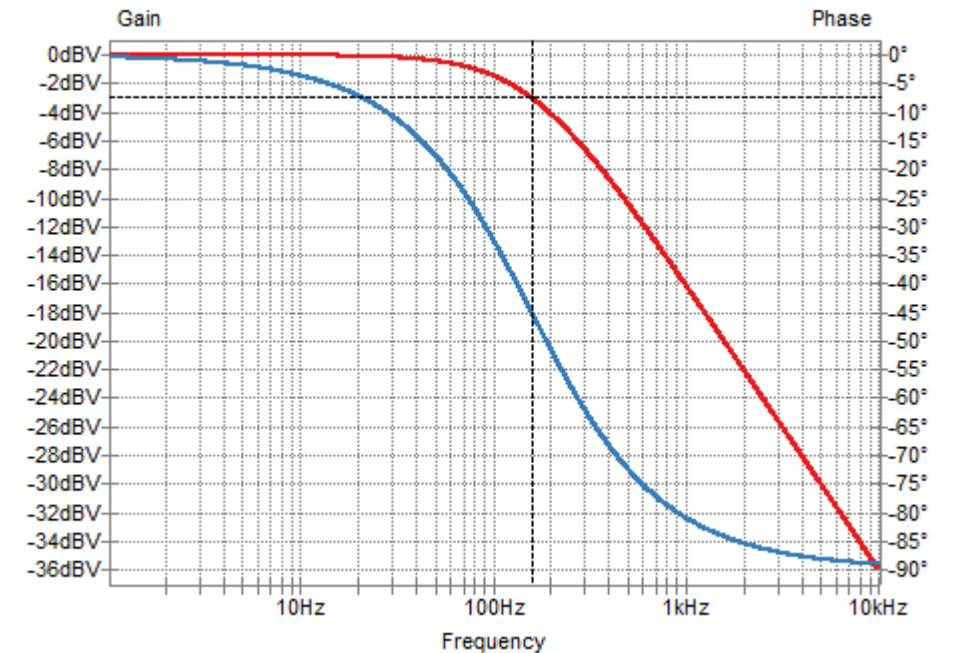
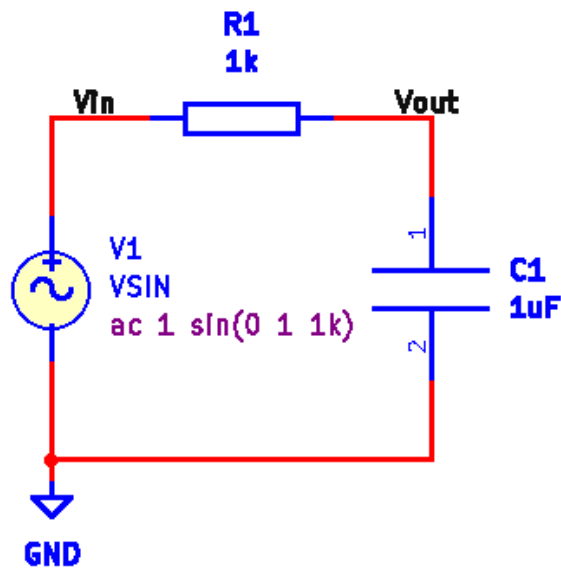
6. With pen and paper, determine the symmetry properties of the amplitude characteristic and the phase characteristic.
7. Plot the magnitude and phase plots in Maple/Python.
8. Notice how different the curves in KiCad/Spice looks. Always take notice of the type of axes used (linear, log, decibel).
9. Use semilogplot to plot the spectra on logarithmic frequency axes in Hz. In this case, plot only positive frequencies. Plot amplitude in decibels.
10. Calculate the inverse Fourier transform of $H(\omega)$ and plot it.
11. Replace the 1k resistor a 500 Ohm resistor. Repeat questions 7 and 10.
12. Compare the two cases and explain using the time scaling property of Fourier transformation.
13. Plot the time delay as a function of frequency in Maple/Python.

Problem 4. RC circuit analysis in frequency domain (sol)

1. Sketch this RC circuit in KiCad/Spice.
2. Run an AC sweep in the frequency range from 1 Hz to 10 kHz and plot magnitude in decibel and phase in degrees.

Spice directive:

.ac dec 1000 1 10k



Problem 4. RC circuit analysis in frequency domain (sol)

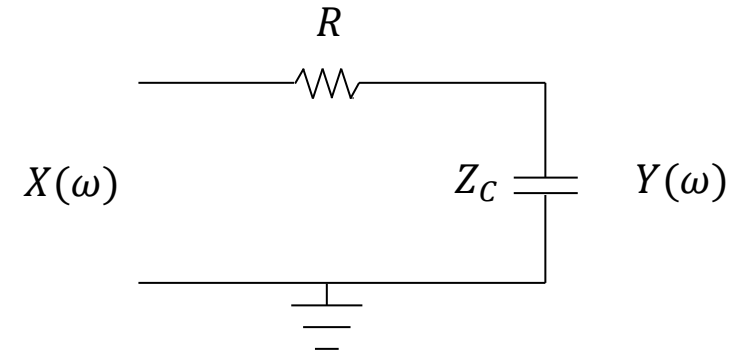
If two resistors in series (R_1, R_2) are set up to divide a total voltage drop of V_{in} , then the voltage output across R_2 is:

$$V_{out} = \frac{R_2}{R_1 + R_2} V_{in}$$

Here R_2 is replaced by the impedance of a capacitor:

$$4: Y(\omega) = \frac{Z_C(\omega)}{R + Z_C(\omega)} X(\omega)$$

Inserting $Z_C(\omega) = \frac{1}{j\omega C}$



$$Y(\omega) = \frac{\frac{1}{j\omega C}}{R + \frac{1}{j\omega C}} X(\omega)$$

$$H(\omega) = \frac{Y(\omega)}{X(\omega)} = \frac{1}{j\omega RC + 1}$$

$$5: H(\omega) = \frac{Y(\omega)}{X(\omega)} = \frac{\frac{1}{RC}}{j\omega + \frac{1}{RC}}$$

Problem 4. RC circuit analysis in frequency domain (sol)

6. With pen and paper, determine the symmetry properties of the amplitude characteristic and the phase characteristic.
7. Plot the magnitude and phase plots in Maple/Python as functions of ω and using linear x- and y axes.
8. Notice how different the curves in KiCad/Spice looks. Always take notice of the type of axes used (linear, log, decibel).
9. Use semilogplot to plot the spectra on logarithmic frequency axes in Hz. In this case, plot only positive frequencies. Plot amplitude in decibels.
10. Calculate the inverse Fourier transform of $H(\omega)$ and plot it.

$$\frac{e^{j\theta_n}}{e^{j\theta_d}} = e^{j(\theta_n - \theta_d)}$$

$$5: H(\omega) = \frac{Y(\omega)}{X(\omega)} = \frac{\frac{1}{RC}}{j\omega + \frac{1}{RC}}$$

$$6: |H(\omega)| = \frac{\left| \frac{1}{RC} \right|}{\left| j\omega + \frac{1}{RC} \right|} = \frac{1}{|j\omega RC + 1|} = \frac{1}{\sqrt{1 + (\omega RC)^2}}$$

$$6: \angle H(\omega) = \angle 1 - \angle(j\omega RC + 1)$$

$$6: \angle H(\omega) = 0 - \tan^{-1} \frac{\omega RC}{1}$$

$$6: \angle H(\omega) = -\tan^{-1} \omega RC$$

Tan and inv Tan are odd functions

Problem 4. RC circuit analysis in frequency domain (sol)

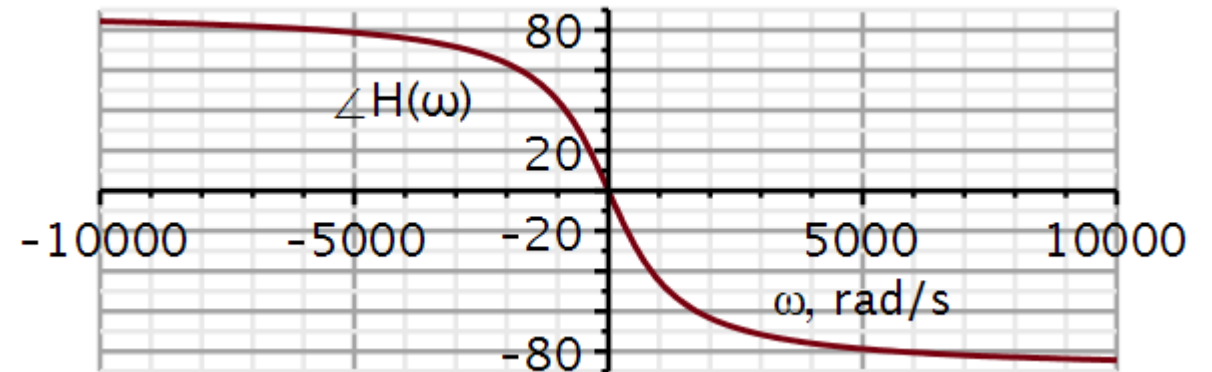
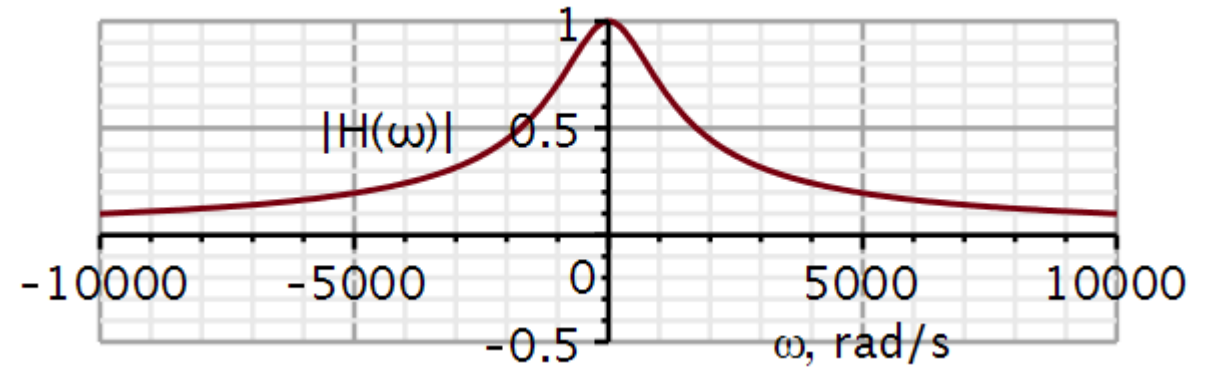
Question 7.

Transfer function

$$H := \omega \rightarrow \frac{1}{j \cdot \omega \cdot R \cdot C + 1} :$$

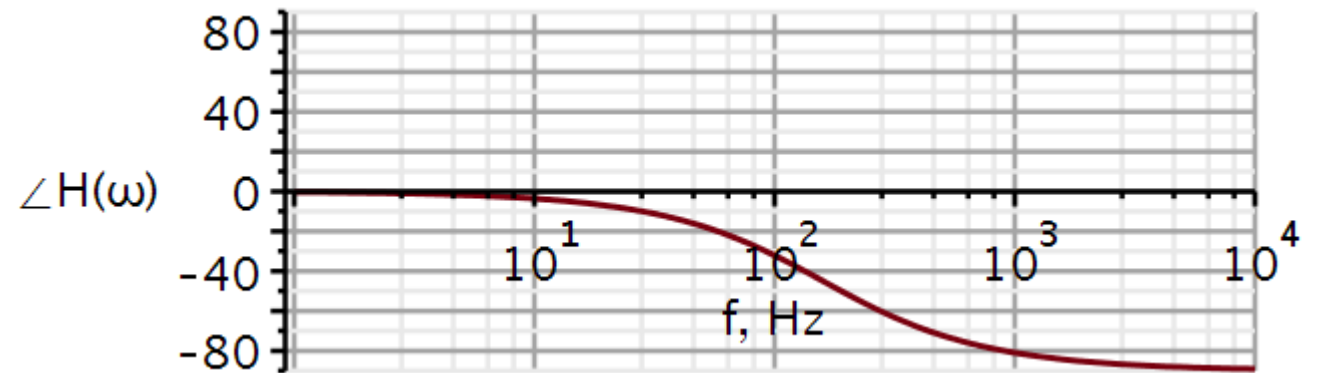
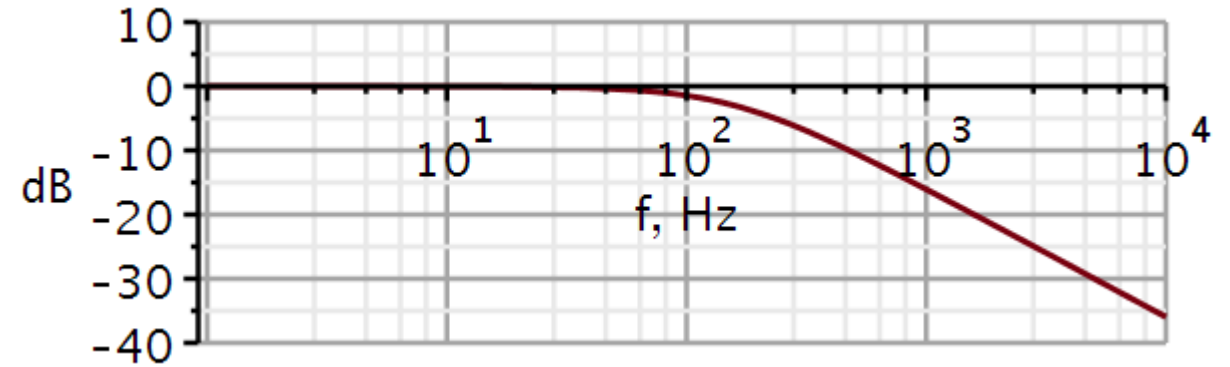
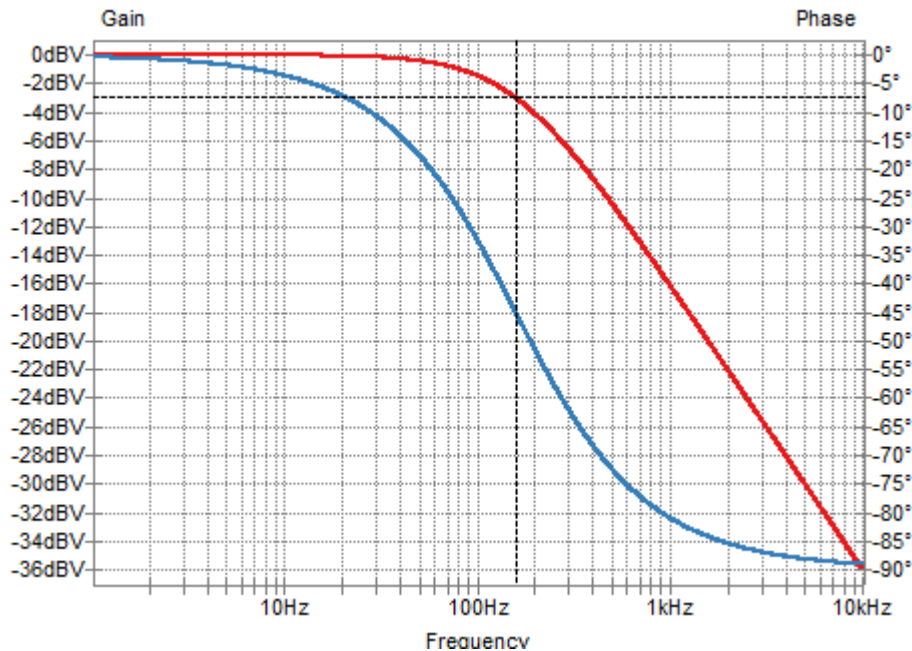
$$R := 1000 ; C := 1\text{E}-6 ;$$

Because we have plotted on linear axes, the curves look very different from those obtained in KiCad/Spice.



Problem 4. RC circuit analysis in frequency domain (sol)

9. Use semilogplot to plot the spectra on logarithmic frequency axes in Hz. In this case, plot only positive frequencies. Plot amplitude in decibels.

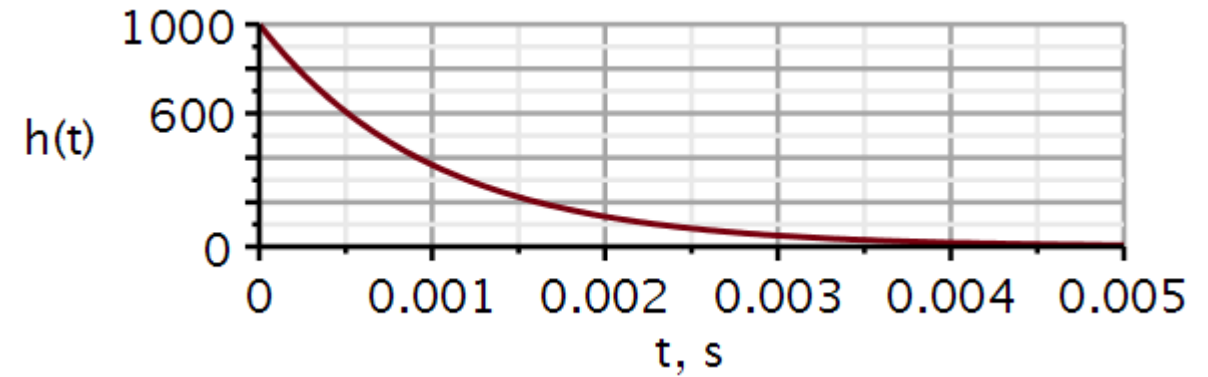


Problem 4. RC circuit analysis in frequency domain (sol)

10. Calculate the inverse Fourier transform of $H(\omega)$ and plot it.

$$h := \text{invfourier}\left(\frac{1}{j \cdot \omega \cdot R \cdot C + 1}, \omega, t\right)$$

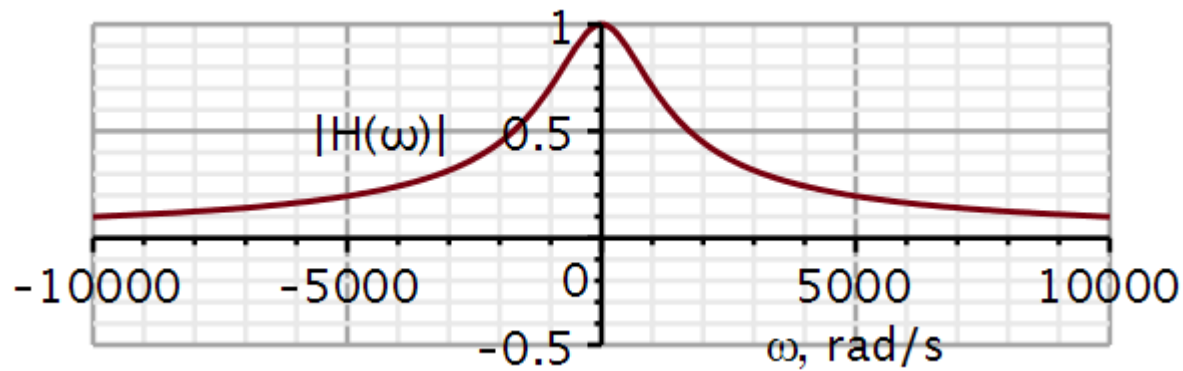
$$h := 1000 \cdot e^{-1000 \cdot t} \text{Heaviside}(t)$$



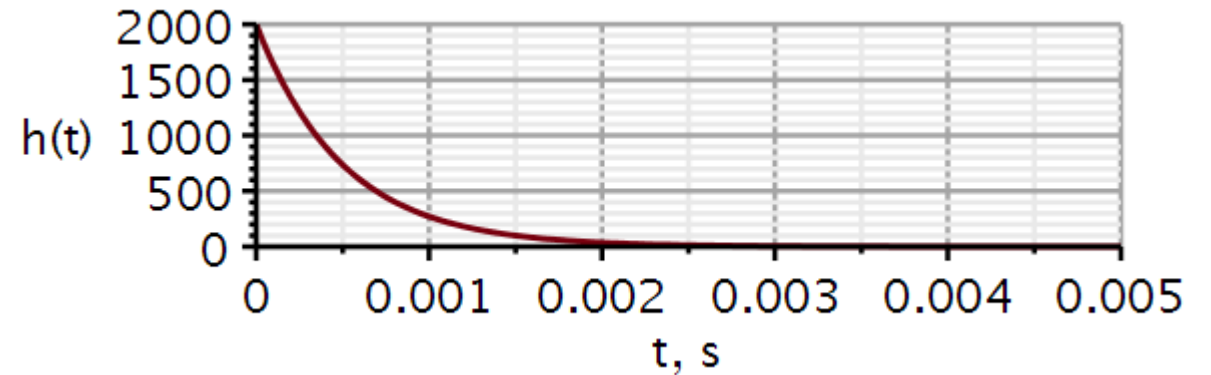
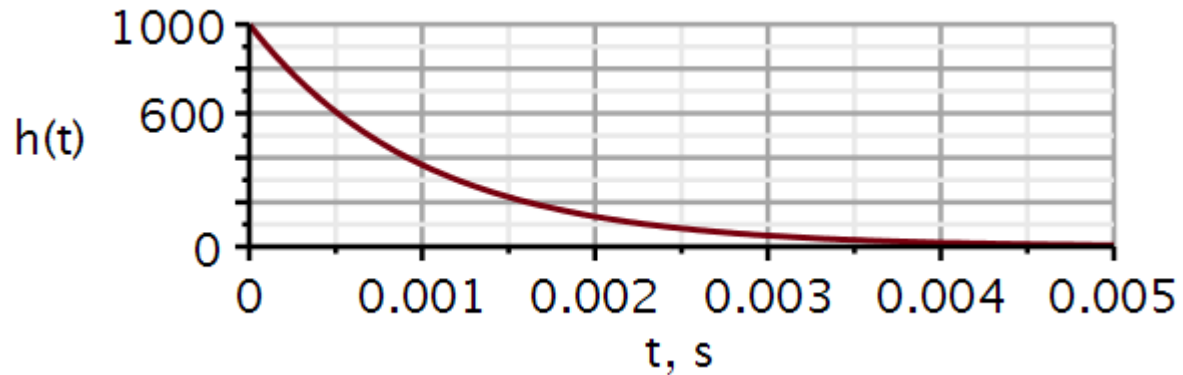
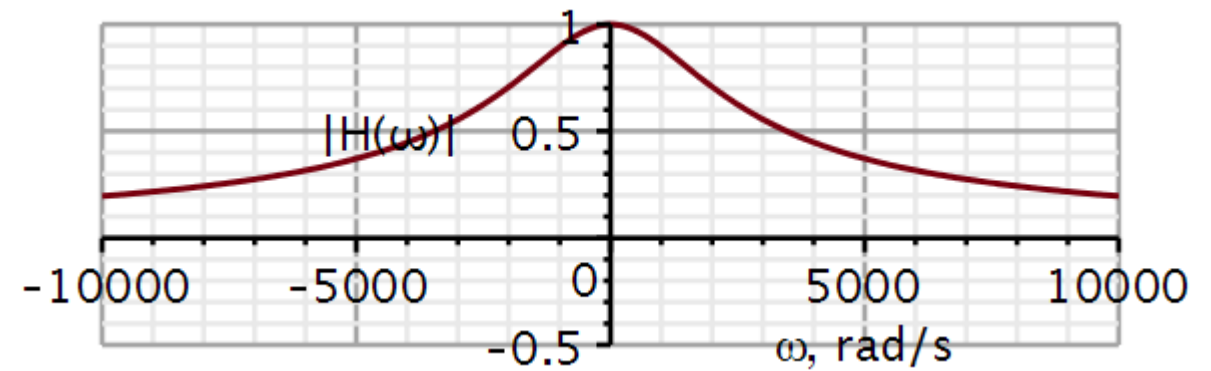
Problem 4. RC circuit analysis in frequency domain (sol)

11. Replace the 1k resistor with a 500 Ohm resistor. Repeat questions 7 and 10.

$$R = 1k\Omega$$



$$R = 500\Omega$$



Problem 4. RC circuit analysis in frequency domain (sol)

12. Compare the two cases and explain using the time scaling property of Fourier transformation.

$$R_0 = 1k\Omega$$

$$R = 500\Omega = \frac{R_0}{2}$$

The impulse response is:

$$h_0(t) = \frac{1}{R_0 C} e^{-t/R_0 C}$$

$$h_1(t) = \frac{2}{R_0 C} e^{-2t/R_0 C}$$

The Fourier transform is:

$$H_0(\omega) = \frac{1}{j\omega R_0 C + 1}$$

$$H_1(\omega) = \frac{1}{j\omega \frac{R_0 C}{2} + 1} = \frac{1}{j\frac{\omega}{2} R_0 C + 1}$$

$$f(at) \leftrightarrow \frac{1}{|a|} F\left(\frac{\omega}{a}\right)$$

$$|a|f(at) \leftrightarrow F\left(\frac{\omega}{a}\right)$$

The time scaling property is very clear. If we scale the time by a factor of 2, then ω is divided by a factor of 2. Hence a wider band width (500 Ω) in the frequency domain yields a shorter lasting response in the time domain, - as illustrated in the previous slide.

Problem 4. RC circuit analysis in frequency domain (sol)

13. Plot the time delay as a function of frequency in Maple/Python.

The derivative of arctan():

$$z := \arctan(a \ t)$$

$$z := \arctan(a \ t)$$

$$\dot{z}$$

$$\frac{a}{a^2 t^2 + 1}$$

At 0 Hz we get a time delay of $RC = 0.5\text{ms}$, i.e., the full effect of the capacitor. As $f \rightarrow \infty$, the capacitor becomes a short circuit, i.e., a wire, and we are left with a resistor circuit with no time delaying properties.

At the cutoff frequency ($\omega = 1/RC$), we get $t_d = \frac{RC}{2} = 0.25\text{ms}$

$$6: \angle H(\omega) = -\tan^{-1} \omega RC$$

$$t_d = -\frac{d}{d\omega} \angle H(\omega) = -\frac{d}{d\omega} (-\tan^{-1} \omega RC)$$

$$t_d = \frac{d}{d\omega} (\tan^{-1} \omega RC) = \frac{1}{(\omega RC)^2 + 1} \cdot \frac{d}{d\omega} (\omega RC)$$

$$t_d = \frac{d}{d\omega} (\tan^{-1} \omega RC) = \frac{RC}{(\omega RC)^2 + 1}$$

